



Addis Ababa University
College of Natural and Computational Sciences
Department of Computer Science

Computer Networking and Security Module

-  **Part I: Data Communication and Computer Networking**
-  **Part II: Network and System Administration**
- Part III: Computer Security**

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CHAPTER 1: INTRODUCTION

DATA & INFORMATION

Data refers to the raw facts that are collected while **information** refers to processed data that enables us to take decisions.

DATA COMMUNICATION

Data Communication is a process of exchanging data or information

1. **Characteristics of Data Communications:-** The effectiveness of any data communications system depends upon the following four fundamental characteristics:
- **Delivery:** The data should be delivered to the correct destination and correct user.
 - **Accuracy:** The communication system should deliver the data accurately, without introducing any errors.
 - **Timeliness:** Audio and Video data has to be delivered in a timely manner without any delay; such a data delivery is called real time transmission of data.
 - **Jitter:** It is the variation in the packet arrival time.

COMPONENTS OF DATA COMMUNICATION

A Data Communication system has five components:-

- **Message:-** is the information to be communicated by the sender to the receiver.
- **Sender** The sender is any device that is capable of sending the data (message).
- **Receiver:** - is a device that the sender wants to communicate the data (message).
- **Transmission Medium:** - It is the path by which the message travels from sender to receiver. It can be wired or wireless and many subtypes in both.
- **Protocol:-** It is an agreed upon set or rules used by the sender and receiver to communicate data. Rules that govern data communication.

DATA REPRESENTATION

Data is collection of raw facts which is processed to deduce information. Data can be represented in: Text, Numbers, Images, Audio and Video.

DATA FLOW

The data can flow between the two devices in the following ways.

- **Simplex:** Only one of the devices sends the data and the other one only receives the data (Unidirectional).
- **Half Duplex:** both the stations can transmit as well as receive but not at the same time.
- **Full duplex mode:** both stations can transmit and receive at the same time.

COMPUTER NETWORK

A collection of an autonomous computing devices connected by a transmission medium. Can be measured by

- Reliability
- Security
- Performance

CATEGORIES OF NETWORK

Networks are categorized on the basis of their size, medium and transmission technology, Architecture ...

Based on size:

- LAN
- MAN
- WAN

Based on Medium,

- Wired vs Wireless

Based on Architecture

- Client Server vs Peer to Peer

Based on Transmission technology

- Packet-switched vs Circuit switched networks

Based on Administration

- Private vs Public

PROTOCOL

Protocol can be defined as set of rules that governs data communications.

It defines **what** is to be communicated, **how** it is to be communicated and **when** it is to be communicated.

There are three key elements of a protocol:

Syntax: It means the structure or format of the data. It is the arrangement of data in a particular order.

Semantics: It tells the meaning of each section of bits and indicates the interpretation of each section.

Timing: It tells the sender about the readiness of the receiver to receive the data

STANDARDS IN NETWORKING

Standards are necessary to ensure interconnectivity and interoperability between various networking hardware and software components or vendors.

Standards in networking are classified into two categories:

- **De facto Standard:** These are the standards that have been traditionally used and mean **by fact** or **by convention**. These standards are not approved by any organized body but are adopted by widespread use.
- **De jure standard:** It means by **law** or **by regulation**. These standards are legislated and approved by a body that is officially recognized.

STANDARD ORGANIZATIONS IN FIELD OF NETWORKING

Standards are created by standards creation committees, forums, and government regulatory agencies.

Examples of Standard Creation Committees

- International Organization for Standardization(ISO)

- International Telecommunications Union – Telecommunications Standard (ITU-T) American National Standards Institute (ANSI)
- Institute of Electrical & Electronics Engineers (IEEE)
- Electronic Industries Associates (EIA)

Examples of

- ATM Forum
- MPLS Forum
- Frame Relay Forum

Examples of Regulatory Agencies:

- Federal Communications Committee (FCC)

2. Review question

CHAPTER TWO: DATA COMMUNICATION

INTRODUCTION

Computer networks are designed to transfer data from one point to another. During transit data is in the form of electromagnetic signals.

DATA & SIGNALS

To be transmitted, data must be transformed to electromagnetic signals.

Data can be Analog or Digital.

- **Analog data** refers to information that is continuous; ex. sounds made by a human voice
- **Digital data** refers to information that has discrete states. Digital data take on discrete values. For example, data are stored in computer memory in the form of 0s and 1s

Signals can be of two types:

- **Analog Signal:** They have infinite values in a range.
- **Digital Signal:** They have limited number of defined values

Periodic & Non Periodic Signals

- Signals which repeat itself after a fixed time period are called **Periodic Signals**.
- Signals which do not repeat itself after a fixed time period are called **Non-Periodic Signals**.
- In data communications, we commonly use periodic analog signals and non-periodic digital signals.

ANALOG SIGNAL

An analog signal has infinitely many levels of intensity over a period of time.

A simple analog signal is a sine wave that cannot be further decomposed into simpler signals.

A sine wave is characterized by three parameters:

- Peak Amplitude
- Frequency
- Phase

CHARACTERISTICS OF AN ANALOG SIGNAL

Peak Amplitude

The amplitude of a signal is the absolute value of its intensity at time t

The peak amplitude of a signal is the absolute value of the highest intensity.

The amplitude of a signal is proportional to the energy carried by the signal

Frequency

Frequency refers to the number of cycles completed by the wave in one second.

Period refers to the time taken by the wave to complete one second.

Phase

Phase describes the position of the waveform with respect to time (specifically relative to time 0). Phase indicates the forward or backward shift of the waveform from the axis

It is measured in degrees or radian

The figure above shows the sine waves with same amplitude and frequency but different phases

Relation between Frequency & Period

- Frequency & Period are inverse of each other.
- It is indicated by the following formula:

$$\begin{array}{c} T = 1/f \\ \text{Or} \\ f = 1/T \end{array}$$

Wavelength

The wavelength of a signal refers to the relationship between frequency (or period) and propagation speed of the wave through a medium.

The wavelength is the distance a signal travels in one period.

It is given by

$$\text{Wavelength} = \text{Propagation Speed} \times \text{Period}$$

OR

$$\text{Wavelength} = \text{Propagation Speed} \times 1/\text{Frequency}$$

It is represented by the symbol: λ (pronounced as lamda)

It is measured in micrometers

It varies from one medium to another.

Composite Signal

A composite signal is a combination of two or more simple sine waves with different frequency, phase and amplitude.

DIGITAL SIGNAL

Digital Signal: Information can also be explained in the form of a digital signal. A digital signal can be explained with the help of following points:

A digital is a signal that has discrete values.

The signal will have value that is not continuous.

LEVEL

Information in a digital signal can be represented in the form of voltage levels.

transmitted has frequency 10Hz.

BIT LENGTH or Bit Interval (T_b)

It is the time required to send one bit.

It is measured in seconds.

BIT RATE

It is the number of bits transmitted in one second.

It is expressed as bits per second (bps).

Relation between bit rate and bit interval can be as follows

Bit rate = 1 / Bit interval

Baud Rate

It is the rate of Signal Speed, i.e the rate at which the signal changes.

A digital signal with two levels 0 & 1 will have the same baud rate and bit rate & bit rate.

TYPES OF CHANNELS

From the point of view of transmission, there are two types of channels:

Low pass Channel

This channel has the lowest frequency as 0 and highest frequency as some non-zero frequency f₁.

This channel can pass all the frequencies in the range 0 to f₁.

Band pass channel

This channel has the lowest frequency as some non-zero frequency f₁ and highest frequency as some non-zero frequency f₂.

This channel can pass all the frequencies in the range f₁ to f₂.

Transmission of Digital signal: Digital signal can be transmitted in the following two ways:

- **Baseband Transmission:** The signal is transmitted without making any change to it (i.e. Without modulation)
 - In baseband transmission, the bandwidth of the signal to be transmitted has to be less than the bandwidth of the channel.
 - Ex. Consider a Baseband channel with lower frequency 0Hz and higher frequency 100Hz, hence its bandwidth is 100 (Bandwidth is calculated by getting the difference between the highest and lowest frequency)
 - A channel whose bandwidth is more than the bandwidth of the signal is called **Wideband channel**
 - A channel whose bandwidth is less than the bandwidth of the signal is called **Narrowband channel**
- **Broadband Transmission:** Given a bandpass channel, a digital signal cannot be transmitted directly through it
 - In **broadband transmission** we use modulation, i.e. we change the signal to analog signal before transmitting it.
 - The digital signal is first converted to an analog signal; since we have a bandpass channel we cannot directly send this signal through the available channel. Ex. Consider the bandpass channel with lower frequency 50Hz and higher frequency 80Hz, and the signal to be

- To pass the analog signal through the bandpass channel, the signal is modulated using a carrier frequency. Ex. The analog signal (10Hz) is modulated by a carrier frequency of 50Hz resulting in a signal of frequency 60Hz which can pass through our bandpass channel.
- The signal is demodulated and again converted into digital signal at the other end.

- Radio waves
- Micro waves
- Infrared.

Wireless signals can be travel or propagated in the following three ways:

TRANSMISSION IMPAIRMENTS & TYPES

Data is transmitted through transmission medium which are not perfect. These imperfections cause signal impairment. Due to the imperfection, error is introduced in the transmitted data i.e. the original signal at the beginning of the transmission is not the same as the signal at the receiver. There are three causes of impairment:

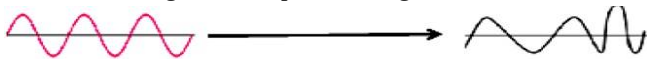
- Attenuation
- Distortion
- Noise

Attenuation

Attenuation results in loss of energy due to distance. When a signal travels through a medium, it loses some of its energy in overcoming the resistance of the medium. The electrical energy in the signal may convert to heat. To compensate for this loss, amplifiers are used to amplify the signal.

Distortion

Distortion changes the shape of the signal.



Noise

Noise is any unwanted signal that is mixed or combined with the original signal during transmission. Due to noise, the original signal is altered and the signal received is not the same as the one sent.

TRANSMISSION MEDIA

Transmission media is a means by which a communication signal is carried from one system to another.

A transmission medium can be defined as anything that can carry information from a source to a destination. The transmission medium is usually **free space, metallic cable or fiber optic cable**. **Guided and Unguided (WIRELESS)** **Guided Transmission Media** uses a cabling system that guides the data signals along a specific path. Out of these, twisted-pair cable, coaxial cable transport signals in the form of electrical signals and fiber-optic cable transport signals in the form of light.

Unguided Transmission Media transport data without using a physical conductor. It uses wireless electromagnetic signals to send data. Three types of unguided media:

- Ground-wave propagation
- Sky-wave propagation
- Line-of-sight propagation

Radio waves: Electromagnetic wave ranging in frequencies between 3 KHz and 1GHz are normally called radio waves. Radio waves are Omni-directional when an antenna transmits radio waves they are propagated in all directions.

Microwaves: Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves.

Microwaves are unidirectional; when an antenna transmits microwaves they can be narrowly focused. Microwaves propagation is line-of-sight.

Infrared: Infrared signals with frequencies ranges from 300 GHz to 400 GHz can be used for short range communication. Infrared signals, having high frequencies, cannot penetrate walls.

Repeater: A repeater is a device that operates only in the physical layer. A repeater receives a signal and, before it

Network Cables - Types

Transmission media carry signals between computers. Can be divided into two broad categories:

Cable-based/Guided media: the signal are guided along a solid medium

Twisted-pair cable: Consists of two conductors (normally copper), each with its own plastic insulation, twisted together.

Two types

- Unshielded Twisted-Pair cable (UTP)
- Shielded Twisted-Pair cable (STP)

Coaxial cable:- uses copper wire to conduct the signals electronically

Fiber-optic cable:- uses glass or plastic conductor and transmits the signals as light

Wireless/Unguided media the signal propagates in the atmosphere and in outer space

NETWORK TOPOLOGY

A *network topology* is a description of the layout of the region or area covered by that network.

There are two types of connections. They are: point-to-point and multi-point.

Point-to-point connections provide a direct link between two devices.

Multi-point connections provide a link between three or more devices on a network.

There are seven basic topologies in the study of network topology:

- Point-to-point topology,
- Bus (point-to-multipoint) topology,
- Ring topology,
- Star topology,
- Hybrid topology,
- Mesh topology and,
- Tree topology.

COMPONENTS OF THE NETWORK/NETWORK DEVICES

becomes too weak or corrupted, regenerates the original bit pattern. A repeater can extend the physical length of a LAN.

A repeater does not actually connect two LANs; it connects two segments of the same LAN. The segments connected are still part of one single LAN. A repeater is not a device that can connect two LANs of different protocols.

HUB: Passive Hubs

A **passive hub** is just a connector. It connects the wires coming from different branches. In a star topology Ethernet LAN, a passive hub is just a point where the signals coming from different stations collide; the hub is the collision point. This type of a hub is part of the media; its location in the Internet model is below the physical layer.

Active Hubs

An **active hub** is actually a multipart repeater. It is normally used to create connections between stations in a physical star topology.

HUB: Hubs are commonly used to connect segments (devices) of a LAN. Hubs don't filter information; instead broadcast incoming packets to all computers connected to it. A hub contains multiple ports which support 8, 12 or 24 RJ-45 ports. Two types: Passive hub and Active hub. It is used in star or ring topology.

SWITCH: A switch filters and forwards data packets across a network.

Unlike a hub which simply replicates what it receives on one port onto all the other ports, a switch keeps a record of the MAC addresses of the devices attached to it.

When the switch receives a data packet, it forwards the packet directly to the recipient device by looking up the MAC address.

- Switch is intelligent and expensive than Hub
- Switch is a point to point communication device
- Hub is a broadcasting device

ROUTER: Router is a specialized network device used to interconnect different types of network that uses different protocol.

Router allows the users to connect several LAN and WAN

- A LAN to another LAN
- A WAN to another WAN
- A LAN to the Internet

Routers use the routing table to determine how to forward packets.

- The list of routing tables provides directions to transfer the data to a particular network destination – various paths to forward the data.
- A router analyzes a destination IP address of a given packet and compares it with the routing table to decide the packet's next best path

A router shares information with other routers in networking.

- Wireless router – offers Wi-Fi connectivity to laptops, smartphones, and other devices with Wi-Fi network capabilities
- Broadband routers / Broadband modem – provided by the internet service provider

(ISP).

BRIDGE: Connects two LANs having the same protocol – (e.g. Ethernet or Token ring).

Filters content by reading the MAC addresses of source and destination.

- MAC to Segment number table
- Do not filter broadcast
- Data not destined for other network is prevented from passing over the bridge
- Slower than repeater due to filtering

GATEWAY: connects networks that use different protocols. Can be:

Transport gateway: connects two computers that use different transport protocols, reformatting packets as need be
Application gateway: understands the format and content of the data and translates messages from one format to another, e.g., an e-mail gateway could translate Internet messages to SMS messages for mobile phones

TYPES OF ERRORS

If the signal comprises of binary data there can be two types of errors which are possible during the transmission:

- **Single bit errors:** In single-bit error, a bit value of 0 changes to bit value 1 or vice versa. Single bit errors are more likely to occur in parallel transmission.
- **Burst Errors:** In Burst error, multiple bits of the binary value changes. Burst error can change any two or more bits in a transmission. Burst errors are more likely to occur in serial transmission.

Redundancy

In order to detect and correct the errors in the data communication we add some extra bits to the original data. These extra bits are nothing but the redundant bits which will be removed by the receiver after receiving the data.

There are different techniques used for transmission error detection and correction.

Detection methods:

- Parity Check
- Cyclic Redundancy Check
- Checksum

SIGNAL ENCODING

Data can be analog or digital, so can be the signal that represents it. Signal encoding is the conversion from analog/digital data to analog / digital signal.

Digital Data to Digital Signal conversion

Coding methods are used to convert digital data into digital signals. There are two types of coding methods:

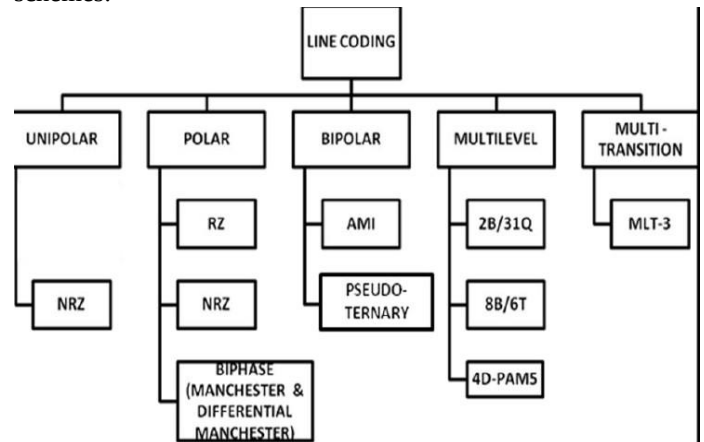
- Line Coding
- Block Coding

Scrambling is also one of the ways to convert digital data to digital signals but is not used.

Line Coding: It is the process of converting Digital data into digital signal.

Classification of Line Codes

The following figure shows the classification of Line coding schemes:



Analog data to analog signal conversion

Modulation: The Process of converting analog data to analog signal is called Modulation. Modulation is used to send an information bearing signal over long distances.

Types of Modulation: Signal modulation can be divided into two broad categories:

Analog modulation and Digital modulation.

- Analog or digital refers to how the data is modulated onto a sine wave. If analog audio data is modulated onto a carrier sine wave, then this is referred to as analog modulation.
- Digital modulation is used to convert digital data to analog signal. Ex ASK, FSK, PSK.

Analog Modulation can be accomplished in three ways:

- Amplitude modulation (AM)
- Frequency modulation (FM)
- Phase modulation (PM).

Digital Modulation Types (Digital to Analog signal conversion)

Digital modulation is used to convert digital data to analog signal. It can be accomplished in the following ways:

- Amplitude Shift Keying(ASK)
- Frequency Shift keying (FSK)
- Phase Shift keying (PSK)

Analog to Digital Conversion using modulation

The definition of the term modulation is described in the next section. Here we have modulation techniques:

- Pulse Amplitude Modulation (PAM)
- Pulse Code Modulation (PCM)
- Pulse Width Modulation (PWM)

CHAPTER 3: NETWORK MODELS

Computer networks are operated by network models; most prominently the OSI reference model and the TCP/ IP Model.

CONCEPT OF LAYERED TASK

The main objective of a computer network is to be able to transfer the data from sender to receiver. This task can be done by breaking it into small sub tasks, each of which is well defined.

Each subtask will have its own process or processes to do. It is called Layers. Every task or job can be done by dividing it into sub task or layers.

INTRODUCTION TO OSI MODEL & ITS LAYERS

The Open Systems Interconnection (OSI) Model was developed by International Organization for Standardization (ISO). ISO is the organization, OSI is the model

It was developed to allow systems with different platforms (Hardware & software) to communicate with each other.

Now, it is a network model that defines the protocols for network communications.

It is a hierarchical model that groups its processes into seven (7) layers. (Top to Bottom)

1. Application Layer
2. Presentation Layer
3. Session Layer
4. Transport Layer
5. Network Layer
6. Data Link Layer
7. Physical Layer

Each layer has specific functions to perform and has to co-operate with the layers above and below it.

DESCRIPTION OF LAYERS IN THE OSI MODEL

Physical Layer:

- The Physical Layer provides a **standardized interface** to physical transmission media, including: Mechanical specification of electrical connectors and cables, for example maximum cable length, Electrical specification of transmission line and Bit-by-bit or symbol-by-symbol delivery.
- **Interface:** The Physical Layer defines the characteristics of interfaces between the devices & transmission medium.
- **Representation of bits:** The physical layer is concerned with transmission of signals from one device to another which involves converting data (1's & 0's) into signals and vice versa. It is not concerned with the meaning or interpretation of bits.
- **Data rate:** The physical layer defines the data transmission rate i.e. number of bits sent per second. It is the responsibility of the physical layer to maintain the defined data rate.
- **Synchronization of bits:** To interpret correct and accurate data the sender and receiver have to maintain the same bit rate and also have synchronized clocks.
- **Line configuration:** The physical layer defines the nature of the connection .i.e. a point to point link, or a multi-point

- **Physical Topology:** The physical layer defines the type of topology in which the device is connected to the network. In a mesh topology it uses a multipoint connection and other topologies it uses a point to point connection to send data.
- **Transmission mode:** The physical layer defines the direction of data transfer between the sender and receiver. Two devices can transfer the data in simplex, half duplex or full duplex mode.
- **Main responsibility of the physical layer:** Transmission of bits from one hop to the next.

Data Link Layer

- The Data Link layer adds reliability to the physical layer by providing error detection and correction mechanisms.
- **Framing:** the Data Link layer receives the data from Network Layer and divides the stream of bits into fixed size manageable units called as **Frames** and sends it to the physical layer. On the receiver side, the data link layer receives the stream of bits from the physical layer and regroups them into frames and sends them to the Network layer.
- **Physical Addressing:** The Data link layer appends the physical address in the header of the frame before sending it to physical layer.
- **Flow control:** The data link layer makes sure that the sender sends the data at a speed at which the receiver can receive it else if there is an overflow at the receiver side the data will be lost.
- **Error Control:** The data link layer imposes error control mechanism to identify lost or damaged frames, duplicate frames and then retransmit them.
- **Access Control:** The data link layer imposes access control mechanism to determine which device has right to send data in an multipoint connection setting.
- **Main Responsibility:** The main responsibility of the data link layer is hop to hop transmission of frames.

Network Layer

- The network layer makes sure that the data is delivered to the receiver despite multiple intermediate devices.
- The network layer at the sending side accepts data from the transport layer, divides it into packets, adds addressing information in the header and passes it to the data link layer.
- The network layer is responsible for source to destination of delivery of data.
- The network layer uses logical address commonly known as IP address to recognize devices on the network.
- Each packet is independent of the other and may travel using different routes to reach the receiver hence may arrive out of turn at the receiver (**Routing**).
- The Network layer does not perform any flow control or error control

- The main responsibility of Network Layer is transmission of packets from source to destination

Transport Layer

- The transport layer takes care of process to process delivery of data and makes sure that it is intact and in order.

- At the sending side, the transport layer receives data from the session layer, divides it into units called segments and sends it to the network layer.
- To ensure process to process delivery the transport layer makes use of **port address** to identify the data from the sending and receiving process.
- The data can be transported in a connection oriented or connectionless manner.
- The Transport layer is responsible for segmentation and reassembly of the message into segments which bear sequence numbers.
- The transport layer also carries out flow control and error control functions; but unlike data link layer these are end to end rather than node to node.
- The **main responsibility** of the transport layer is process to process delivery of the entire message

2. Internet Layer
3. Transport Layer

Session Layer

- The session layer establishes a session between the communicating devices called **dialog** and **synchronizes** their interaction. It is the responsibility of the session layer to establish and synchronize the dialogs. It is also called the network **dialog controller**.
- The session layer at the sending side accepts data from the presentation layer adds checkpoints to it called **sync** bits and passes the data to the transport layer. At the receiving end the session layer receives data from the transport layer removes the checkpoints inserted previously and passes the data to the presentation layer. So, the **main responsibility of session layer** is dialog control and synchronization.

Presentation Layer

- The communicating devices may be having different platforms. The presentation layer performs translation, encryption and compression of data.
- The presentation layer receives the data from the application layer adds header which contains information related to encryption and compression and sends it to the session layer.
- So, the main responsibility of the Presentation layer is translation, compression and encryption.

Application Layer

- The application layer enables the user to communicate its data to the receiver by providing certain services.
- The main Responsibility of Application layer is to provide the user access to network resources.

TCP/IP MODEL, ADDRESSING IN TCP/IP – IPV4

After understanding the concept of layered task and the OSI model, we introduce the TCP/IP model. This model is currently being used on our systems. TCP/IP model is a collection of protocols often called **a protocol suite**. It offers a rich variety of protocols.

It is a hierarchical model; it existed even before the OSI model was developed. Originally had four layers (bottom to top):

1. Host to Network Layer

4. Application Layer

The structure TCP/IP model is very similar to the structure of the OSI reference model. The OSI model has seven layers where the TCP/IP model has four layers.

- The Application layer of TCP/IP model corresponds to the Application Layer of Session, Presentation & Application Layer of OSI model.
- The Transport layer of TCP/IP model corresponds to the Transport Layer of OSI model
- The Network layer of TCP/IP model corresponds to the Network Layer of OSI model
- The Host to network layer of TCP/IP model corresponds to the Physical and Data link Layer of OSI model.

FUNCTIONS OF THE LAYERS OF TCP/IP MODEL

Host to Network Layer

This layer is a combination of protocols at the physical and data link layers. It supports all standard protocols used at these layers.

Network Layer or IP

Also called as the Internetwork Layer (IP). It holds the IP protocol which is a network layer protocol and is responsible for source to destination transmission of data.

The Internetworking Protocol (IP) is a **connection-less & unreliable protocol**.

It is a best effort delivery service. i.e. there is no error checking in IP, it simply sends the data and relies on its underlying layers to get the data transmitted to the destination. Even if it is unreliable IP cannot be considered weak and useless; since it provides only the functionality that is required for transmitting data thereby giving maximum efficiency.

Since there is no mechanism of error detection or correction in IP, there will be no delay introduced on a medium where there is no error at all.

IP is a combination of four protocols:

1. ARP
2. RARP
3. ICMP
4. IGMP

ARP – Address Resolution Protocol: It is used to resolve the physical address of a device on a network, where its logical address is known.

Physical address is the 48 bit address that is imprinted on the NIC or LAN card, Logical address is the Internet Address or commonly known as IP address that is used to uniquely & universally identify a device.

RARP– Reverse Address Resolution Protocol: It is used by a device on the network to find its Internet address when it knows its physical address.

ICMP- Internet Control Message Protocol: It is a signaling mechanism used to inform the sender about datagram problems that occur during transit. It is used by

intermediate devices.

IGMP- Internet Group Message Protocol: It is a mechanism that allows sending the same message to a group of recipients.

Transport Layer

Transport layer protocols are responsible for transmission of data running on a process of one machine to the correct process running on another machine.

The transport layer contains three protocols:

1. TCP
2. UDP
3. SCTP

TCP – Transmission Control Protocol: TCP is a reliable connection-oriented, reliable protocol. i.e. a connection is established between the sender and receiver before the data can be transmitted. It divides the data it receives from the upper layer into segments and tags a sequence number to each segment which is used at the receiving end for reordering of data.

UDP – User Datagram Protocol: UDP is a simple protocol used for process to process transmission. It is an unreliable, connectionless protocol for applications that do not require flow control or error control. It simply adds port address, checksum and length information to the data it receives from the upper layer.

SCTP – Stream Control Transmission Protocol: SCTP is a relatively new protocol added to the transport layer of TCP/IP protocol suite. It combines the features of TCP and UDP. It is used in applications like voice over Internet and has a much broader range of applications

Application Layer

The Application Layer is a combination of Session, Presentation & Application Layers of OSI models and defines high level protocols like File Transfer (FTP), Electronic Mail (SMTP), Virtual Terminal (TELNET), Domain Name Service (DNS), etc.

IP ADDRESSING

Packets in the IPv4 format are called datagram. An IP datagram consists of a header part and a text part (payload).

IPv4 can be explained with the help of following points:

- IP addresses
- Address Space
- Notations used to express IP address
- Classful Addressing
- Subnetting
- CIDR
- NAT
- IPv4 Header Format

IP addresses: IP address uniquely identifies a device on the Internet. Every host and router on the Internet has an IP address, which encodes its network number and host number.

- The combination is unique: in principle, no two machines on the Internet have the same IP address.
- An IPv4 address is 32 bits long
- They are used in the Source address and Destination address fields of IP packets.

- An IP address does not refer to a host but it refers to a network interface.

Address Space: An address space is the total number of addresses used by the protocol.

- If a protocol uses N bits to define an address, the address space is 2^N because each bit can have two different values (0 or 1) and N bits can have 2^N values.
- IPv4 uses 32-bit addresses, which means that the address space is 2^{32} or 4,294,967,296 (more than 4 billion).

Notations: There are two notations to show an IPv4 address:

- Binary notation: The IPv4 address is displayed as 32 bits. ex. 11000001 10000011 00011011 11111111
- Dotted decimal notation: To make the IPv4 address easier to read, Internet addresses are usually written in decimal form with a decimal point (dot) separating the bytes. Each byte (octet) is 8 bits hence each number in dotted-decimal notation is a value ranging from 0 to 255. Ex. 129.11.11.239

Classful addressing : In classful addressing, the address space is divided into five classes: A, B, C, D, and E.

- Netid and Hostid In classful addressing, an IP address in class A, B, or C is divided into netid and hostid.

Subnetting: It allows a network to be split into several parts for internal use but still act like a single network to the outside world.

- To implement subnetting, the router needs a subnet mask that indicates the split between network + subnet number and host. Ex. 255.255.252.0/22. A/22 to indicate that the subnet mask is 22 bits long.

CIDR: A class B address is far too large for most organizations and a class C network, with 256 addresses is too small. This leads to granting Class B address to organizations who do not require all. This is resulting in depletion of Address space. A solution is CIDR (Classless InterDomain Routing) The basic idea behind CIDR, is to allocate the remaining IP addresses in variable-sized blocks, without regard to the classes.

NAT (Network Address Translation): The scarcity of network addresses in IPv4 led to the development of IPv6. Due to this, we need to use private IP address inside the organization and translate it to public IP address using NAT.

IP Header: IPv4 has 32 bit header information such as Version, IHL, total length, types of services

- There are two kinds of routing algorithms (routing tables): nonadaptive (static) and adaptive (dynamic)
- And two kinds of protocols
 - Interior routing protocols
 - Exterior routing protocols

ROUTING AND ROUTING PROTOCOLS

Routing is the act of moving information across an internetwork from a source to a destination. At the heart of such protocol is the routing algorithm that determines the path for a packet.

- Routing requires a host or a router to have a routing table which is constructed by the routing algorithm.

CHAPTER FOUR: SWITCHING AND MULTIPLEXING

line wastes the available bandwidth. Multiplexing allows us to combine several lines into one.

SWITCHING

Switching is a mechanism by which data/information sent from source towards destination which are not directly connected. Networks have interconnecting devices, which receives data from directly connected sources, stores data, analyze it and then forwards to the next interconnecting device closest to the destination.

Switching can be categorized as:

Circuit Switching: In circuit switching, two communicating stations are connected by a dedicated communication path which consists of intermediate nodes in the network and the links that connect these nodes.

It involved the following three distinct steps

- **Circuit Establishment:** To establish an end-to-end connection before any transfer of data.
- **Data transfer:** Transfer data is from the source to the destination. The data may be analog or digital, depending on the nature of the network. The connection is generally full-duplex.
- **Circuit disconnect:** Terminate connection at the end of data transfer. Signals must be propagated to deallocate the dedicated resources.

Message Switching:

Packet Switching: Packet switching was designed to address the shortcomings of circuit switching in dealing with data communication. Unlike circuit switching where communication is continuous along a dedicated circuit, in packet switching, communication is discrete in form of packets. The three switching modes to choose from are:

- Cut-through
- Store-and-forward
- Fragment-free

MULTIPLEXING

Whenever the bandwidth of a medium linking two devices is greater than the bandwidth needs of the devices, the link can be shared. Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link.

Frequency division Multiplexing: Frequency-division multiplexing (FDM) is an analog technique that can be applied when the bandwidth of a link (in hertz) is greater than the combined bandwidths of the signals to be transmitted.

In FDM, signals generated by each sending device modulate different carrier frequencies.

Wavelength-division multiplexing (WDM): is designed to use the high-data-rate capability of fiber-optic cable. The optical fiber data rate is higher than the data rate of metallic transmission cable. Using a fiber-optic cable for one single

Time division multiplexing: different ongoing connections can use same switching path but at different interleaved time intervals.). Each connection occupies a portion of time in the link.

MEDIUM ACCESS CONTROL

The problem of controlling the access to the medium is similar to the rules of speaking in a meeting (the right to speak is upheld; two people do not speak at the same time; do not interrupt each other; do not monopolize the discussion; ...) Three categories of multiple access protocols

Random Access Protocols - try your best like taxis do

- MA - Multiple Access
- CSMA - Carrier Sense MA
- CSMA/CD - CSMA with Collision Detection
- CSMA/CA - CSMA with Collision

Avoidance Controlled-Access Protocols - get permission

- Reservation
- Polling
- Token Passing

Channelization Protocols - simultaneous use

- FDMA - Frequency-Division MA
- TDMA - Time-Division MA
- CDMA - Code-Division MA

main classes namely – system software and application software.

Network Overview

MODULE II: SYSTEM AND NETWORK ADMINISTRATION

CHAPTER 1: INTRODUCTION AND BACKGROUND

Computer Systems and Network overview

Computer System

Computer system is a collection of entities (hardware and software) that are designed to receive process, manage and present information in a meaningful format. Hardware refers to the physical, tangible computer equipment and devices, which provide support for major functions such as input, processing (internal storage, computation and control), output, secondary storage (for data and programs), and communication. There are five main hardware components in a computer system: Input, Processing, Storage, Output and Communication devices. Computer software, also known as programs or applications, are the intangible components of the computer system. They can be classified into two

A network can be defined as two or more computers connected together in such a way that they can share resources. The primary purpose of a network is to share resources, and a resource could be:

- a file,
- a folder,
- a printer,
- a disk drive, or just about anything else that exists on a computer.

Therefore, a computer network is simply a collection of computers or other hardware devices that are connected together, either physically or logically, using special hardware and software, to allow them to exchange information and cooperate. Networking is the term that describes the processes involved in designing, implementing, upgrading, managing and otherwise working with networks and network technologies.

There are different types of a computer networks based on their respective attributes. These includes: geographical span, inter-connectivity (physical topology), administration and architecture.

Geographical Span: based on geographical area it covers there are different types of network:

Personal Area Network (PAN): is a network may be spanned across a given table with distances between the devices not more than few meters. The technology used to interconnect the devices could be a Bluetooth. These networks are called Personal Area Networks, since the devices interconnected in these networks are belongs to a single person.

Local Area Network (LAN): is a network that may span across a building, or across several buildings within a single organization, by using intermediate devices, like switches and/or hubs, to interconnect devices in all floors.

Metropolitan Area Network (MAN): is a network that may span across a whole city interconnecting several buildings and organizations.

Wide Area Network (WAN): is a network that may span across multiple cities, or an entire country, or an entire continent or it may even cover the whole world. For example, an Internet is one example of WAN.

Inter-connectivity: components of a network, including end devices and interconnecting devices, can be connected to each other differently in some fashion. By connectedness we mean either logically, physically or both ways. Network topology refers to the shape of a network, or the network's layout. It is

the geometric representation of the relationship of all the links and linking devices to one another. There are

four basic types of topologies, namely **bus**, **star**, **ring** and **mesh** topologies.

and it can be calculated by using the general formula $((n(n-1))/2)$, where n is the number of nodes in the

Bus Topology: in this topology all devices are connected to a central cable, called the bus or backbone, which is terminated at its ends. The purpose of the terminators is to stop the signal from bouncing, thereby clearing the cable so that other computers can send data. Message transmitted along the Bus is visible to all computers connected to the backbone cable. As the message arrives at each workstation, the workstation checks the destination address contained in the message either to process or drop the packet if it matches or not respectively. Its advantages are, ease of installation and less amount of cable requirement. Its main drawback is, the entire network will be shut down if there is a break in the main cable.

Star Topology: in this topology, each node is connected directly to a central device called a hub or a switch. Data on a star network passes through the central device (switch) before continuing to its destination. The central device manages and controls all functions of the network. This configuration is common with twisted pair cable. RJ-45 Connectors are used to connect the cable to the Network Interface Card (NIC) of each computer. Its advantages include, ease of installation and reconfiguration, robust (ease of fault identification and isolation), link failure only affects device(s) connected to that link, and is less expensive than mesh. Its drawbacks include more cable requirements (than bus and ring) and single point of failure (if central device fail, the whole system will be down).

Ring Topology: in this topology, all devices are connected to one another in the shape of a closed loop, so that each device is connected directly to two other devices, one on either side of it. Some of its advantages include, easy to install and reconfigure, less expensive (than mesh), and performance is even despite the number of users. Its cons include, break in the ring (such as a disabled station) can disable the entire network, and limitations on media and traffic (limitation on ring length and number of devices).

Mesh Topology: in this topology devices are connected with many redundant interconnections between network nodes. In a full mesh topology, every node has a connection to every other node in the network, which makes it the most expensive topology over all the other topologies. The number of cables grows fast as the number of nodes increases,

network. It has several benefits, such as: dedicated links between devices, robust (single link failure don't affect entire network), privacy/security (direct communication between communicating devices), and ease of fault identification and isolation. Its drawbacks include, installation and reconnection are difficult (large number of cables), huge amount of cables consumes a lot of space, and it is the most expensive of all.

Hybrid Topology: A network structure whose design contains more than one topology is said to be Hybrid Topology. Hybrid topology inherits merits and demerits of all the incorporating topologies. As its name indicates, this topology can be created by merging one or more of the above basic topologies.

Administration: From administrator's point of view, a network can be **private network** which belongs to a single autonomous system and cannot access outside of its physical or logical domain. Or a network can be **public network**, which can be accessed by anyone inside or outside of an organization.

Network Architecture: based on the architecture (where do the clients get the shared resources?), networks can be categorized into three:

Client-Server Architecture: There can be one or more systems acting as Server. Other being Client, request the Server to serve requests. Servers take and process request on clients' behalf.

Peer-to-Peer (Point-to-point): Two systems can be connected Point-to-Point, or in other words back-to-back fashion. They both reside on same level and called peers.

There can be hybrid network which involves network architecture of both the above types.

Network Protocols

Protocol is a set of rules or standards that control data transmission and other interactions between networks, computers, peripheral devices, and operating systems.

While to devices communicate with each other, the same protocol must be used on the sending and receiving devices. It is possible for two devices that use different protocols to communicate with each other, but a gateway is needed in between.

The TCP/IP protocol suite was developed prior to the OSI model. Therefore, the layers in the TCP/IP protocol suite do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: host-to-network, Internet, transport, and application layers. However, when TCP/IP is compared to OSI, we can say that the host-to-network layer is equivalent to the combination of the physical and data link layers. The Internet layer is equivalent to the network layer, and the application layer is roughly doing the job of the session, presentation, and application layers with the transport layer in TCP/IP taking care of part of the duties of the session layer.

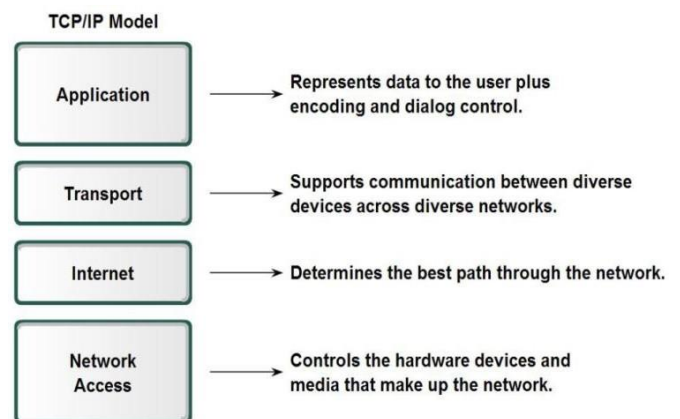
TCP/IP is a hierarchical protocol made up of interactive modules, each of which provides a specific functionality; however, the modules are not necessarily interdependent. Whereas the OSI model specifies which functions belong to each of its layers, the layers of TCP/IP suite contain relatively independent protocols that can be mixed and matched depending on the needs of the system. The term hierarchical means that each upper-level protocol is supported by one or more lower-level protocols.

At the transport layer, TCP/IP defines three protocols: Transmission Control Protocol (TCP), User Datagram Protocol (UDP), and Stream Control Transmission Protocol (SCTP). At the network layer, the main protocol defined by TCP/IP is the Internetworking Protocol (IP); there are also some other protocols that support data movement in this layer.

Figure. TCP/IP Protocol Stack

Network Access (Physical and Data Link Layers)

The Network Access layer of the TCP/IP model corresponds with the Data Link and Physical layers of the OSI reference model. It defines the protocols and hardware required to connect a host to a physical network and to deliver data across it. Packets from the Internet layer are sent down



the Network Access layer for delivery within the physical network. The destination can be another host in the network, itself, or a router for further forwarding. So the Internet layer has a view of the entire Internetwork whereas the Network Access layer is limited to the physical layer boundary that is often defined by a layer 3 device such as a router.

The Network Interface layer (also called the Network Access layer) is responsible for placing TCP/IP packets on the network medium and receiving TCP/IP packets off the network medium. TCP/IP was designed to be independent of the network access method, frame format, and medium. In this way, TCP/IP can be used to connect differing network types. These include LAN technologies such as Ethernet and Token Ring and WAN technologies such as X.25 and Frame Relay. Independence from any specific network technology gives TCP/IP the ability to be adapted to new technologies such as Asynchronous Transfer Mode (ATM).

Network Access layer uses a physical address to identify hosts and to deliver data.

The Network Access layer PDU is called a frame. It contains the IP packet as well as a protocol header and trailer from this layer.

The Network Access layer header and trailer are only relevant in the physical network. When a router receives a frame, it strips of the header and trailer and adds a new header and trailer before sending it out the next physical network towards the destination.

which is transported separately. Datagrams can travel along different routes and can arrive out of sequence or be

Network (Internet) Layer

At the network layer (or, more accurately, the Internetwork layer), TCP/IP supports the Internetworking Protocol. IP, in turn, uses four supporting protocols: ARP, RARP, ICMP, and IGMP.

The Internet (Network) Layer Protocols

Internet Protocol (IP): IP essentially is the Internet layer. The other protocols found here merely exist to support it. It is an unreliable and connectionless protocol (i.e. a best-effort delivery service). The term best effort means that IP provides no error checking or tracking. It assumes the unreliability of the underlying layers and does its best to get a transmission through to its destination, but with no guarantees.

IP transports data in packets called datagrams, each of

duplicated. IP does not keep track of the routes and has no facility for reordering datagrams once they arrive at their destination.

Internet Control Message Protocol (ICMP): works at the Network layer and is used by IP for many different services. ICMP is a management protocol and messaging service provider for IP. The following are some common events and messages that ICMP relates to:

Destination Unreachable: If a router can't send an IP datagram any further, it uses ICMP to send a message back to the sender, advising it of the situation.

Buffer Full: If a router's memory buffer for receiving incoming datagrams is full, it will use ICMP to send out this message until the congestion abates.

Hops: Each IP datagram is allotted a certain number of routers, called hops, to pass through. If it reaches its limit of hops before arriving at its destination, the last router to receive that datagram deletes it. The executioner router then uses ICMP to send an obituary message, informing the sending machine of the demise of its datagram.

Ping (Packet Internet Groper) uses ICMP echo messages to check the physical and logical connectivity of machines on a network.

Traceroute Using ICMP timeouts, Traceroute is used to discover the path a packet takes as it traverses an Internet network.

Address Resolution Protocol (ARP): finds the hardware address (physical or MAC address) of a host from a known IP address. ARP interrogates the local network by sending out a broadcast asking the machine with the specified IP address to reply with its hardware address. **Reverse Address**

Resolution Protocol

(RARP): discovers the identity of the IP address for diskless machines by sending out a packet that includes its MAC address and a request for the

IP address assigned to that MAC address. A designated machine, called a RARP server, responds with the answer, and the identity crisis is over.

Transport Layer

Traditionally the transport layer was represented in TCP/IP by two protocols: TCP and UDP. IP is a host-to-host protocol, meaning that it can deliver a packet from one physical device to another. UDP and TCP are transport level protocols responsible for

delivery of a message from a process (running program) to another process. A new transport layer protocol, SCTP, has been devised to meet the needs of some newer applications.

The Transport Layer Protocol

- Transmission Control Protocol (TCP)
- User Datagram Protocol (UDP)
- Stream Control Transmission Protocol (SCTP)

The Port Numbers

TCP and UDP must use port numbers to communicate with the upper layers, because they're what keep track of different conversations crossing the network simultaneously. These port numbers identify the source and destination application or process in the TCP segment. There are $2^{16} = 65,536$ ports available.

Well-known ports: The port numbers range from 0 to 1,023.

Registered ports: The port numbers range from 1,024 to 49,151. Registered ports are used by applications or services that need to have consistent port assignments.

Dynamic or private ports: The port numbers range from 49,152 to 65,535.

TCP Ports		UDP Ports	
Protocol	TCP Port Number	Protocol	TCP Port Number
Telnet	23	SNMP	161
SMTP	25	TFTP	69
HTTP	80	DNS	53
FTP	21	POP3	110
DNS	53	DHCP	68
HTTPS	443	NTP	123
SSH	22	RPC	530

Table. Sample TCP and UDP port numbers from well-known category

Application Layer

The application layer in TCP/IP is equivalent to the combined session, presentation, and application layers in the OSI model and many protocols are defined at this layer.

The Process/Application Layer Protocols

- Telnet
- File Transfer Protocol (FTP)
- Network File System (NFS).
- Simple Mail Transfer Protocol (SMTP)
- POP3 is used to receive mail.
- Simple Network Management Protocol (SNMP)
- Domain Name Service (DNS)
- Dynamic Host Configuration Protocol (DHCP)

Philosophy of System Administration

Network and system administration is a branch of engineering that concerns the operational management of human-computer systems. It is about putting together a network of computers (workstations, PCs and supercomputers), getting them running and then keeping

them running in spite of the activities of users who tend to cause the systems to fail.

A system administrator works for users, so that they can use the system to produce work.

However, a system administrator should not just cater for one or two selfish needs, but also work for the benefit of a whole community. Today, that community is a global community of machines and organizations, which spans every niche of human society and culture, thanks to the Internet. It is often a difficult balancing act to determine the best policy, which accounts for the different needs of everyone with a stake in a system. Once a computer is attached to the Internet, we have to consider the consequences of being directly connected to all the other computers in the world.

It's about hardware, software, user support, diagnosis, repair and prevention. System administrators need to know a bit of everything: the skills are technical, administrative and socio-psychological.

The terms network administration and system administration exist separately and are used both variously and inconsistently by industry and by academics.

System administration is the term used traditionally by mainframe and Unix engineers to describe the management of computers whether they are coupled by a network or not. To this community, network administration means the management of network infrastructure devices (routers and switches). The world of personal computers (PCs) has no tradition of managing individual computers and their subsystems, and thus does not speak of system administration, per se. To this community, network administration is the management of PCs in a network. In this material, we shall take the first view, since this is more precise.

Network and system administration are increasingly challenging. The complexity of computer systems is increasing all the time. Even a single PC today, running Windows NT, and attached to a network, approaches the level of complexity that mainframe computers had ten years ago.

We are now forced to think systems not just computers.

Scope of Network administration

The management of a network, usually called network administration, consists of procedures and services that keep the network running properly. An important part of network management entails making sure that the network is available (or up and running as IT professionals say) when employees and managers need it. Other admin activities are:

- Monitoring the network capacity to ensure

that all transmission requirements can be met.

- Adding capacity to the network by increasing bandwidth, interconnecting additional nodes, or creating and interconnecting additional networks.

- Training people to use the network effectively
- Assisting IT professionals in organizational applications that will make good use of the network's capabilities.
- Backing up the network software and data regularly to protect against the failure of network or any of its components
- Putting security procedures in place to make certain that only authorized users have access to the network and ensuring that all security procedures are followed
- Making sure the network personnel can respond quickly and effectively in the event of a network operational or security failure.
- Diagnosing and troubleshooting problems on the network and determining the best course of action to take to solve them.
- Planning and implementing adequate security.
- Providing a comfortable environment for users.

Although the specifics of being a system administrator may change from platform to platform, there are underlying themes that do not. These themes make up the philosophy of system administration. The themes are:

- Automate everything
- Document everything
- Communicate as much as possible
- Know your resources
- Know your users
- Know your business
- Security cannot be an afterthought
- Plan ahead
- Expect the unexpected
- Backup and disaster recovery planning
- Patching

The goal of Network administration

The goal is to keep the network running properly and configuring and managing services that are provided over the network.

There are many services that we use regularly. There are some which work in the background enabling other services to run smoothly.

The challenges of system administration

System administration is not just about installing operating systems. It is about planning and designing an efficient community of computers so that real users will be able to get their jobs done. That means:

- Designing a network which is logical and efficient.
- Deploying large numbers of machines which can be easily upgraded later.
- Deciding what services are needed.

- Developing ways of fixing errors and problems which occur.
- Keeping track of and understanding how to use the enormous amount of knowledge which increases every year.

Some system administrators are responsible for both the hardware of the network and the computers which it connects, i.e. the cables as well as the computers. Some are only responsible for the computers. Either way, an understanding of how data flow from machine to machine is essential as well as an understanding of how each machine affects every other.

The Meta principles of system administration

Principle 1 (Policy is the foundation)

System administration begins with a policy – a decision about what we want and what should be, in relation to what we can afford.

Policy speaks of what we wish to accomplish with the system, and what we are willing to tolerate of behavior within it. It must refer to both the component parts and to the environment with which the system interacts. If we cannot secure predictability, then we cannot expect long-term conformance with a policy.

Principle 2 (Predictability)

The highest level aim in system administration is to work towards a predictable system. Predictability has limits. It is the basis of reliability, hence trust and therefore security.

Policy and predictability are intertwined. What makes system administration difficult is that it involves a kind of ‘search’ problem. It is the hunt for a stable region in the landscape of all policies, i.e. those policies that can lead to stable and predictable behavior. In choosing policy, one might easily promote a regime of cascading failure, of increasing unpredictability that degenerates into chaos. Avoiding these regimes is what makes system administration difficult. As networks of computers and people grow, their

interactions become increasingly complex and they become non-deterministic, i.e. not predictable in terms of any manageable number of variables. We therefore face another challenge that is posed by inevitable growth.

Principle 3 (Scalability)

Scalable systems are those that grow in accordance with policy; i.e. they continue to function predictably, even as they increase in size.

Human-computer systems are too complex and have too many interactions and dependencies to be deterministic

When we speak of predictability, it must always be within a margin of error. If this were not the case, system administration would not be difficult

named WORKGROUP (MSHOME in Windows XP). To access shared resources on other PCs within its group, a

CHAPTER 2: WINDOWS NETWORK CONCEPTS

In computer networking a workgroup is a collection of computers on a LAN that share common resources and responsibilities. Workgroup is Microsoft's term for a peer-to-peer L. Windows WORKGROUPs can be found in homes, schools and small businesses. Computers running Windows OSs in the same work group may share files, printers, or Internet connection. Workgroup contrasts with domain, in which computers rely on centralized authentication.

Windows workgroup

A **Windows workgroup** is a group of standalone computers in a peer-to-peer network. Each computer in the workgroup uses its own local accounts database to authenticate resource access. The computers in a workgroup also do not have a common authentication process. The default-networking environment for a clean windows load is workgroup

In general, a given Windows workgroup environment can contain many computers but work best with 15 or fewer computers. As the number of computers increases, a workgroup eventually become very difficult to administer and should be re-organized into multiple networks or set up as a client-server network.

The computers in a workgroup are considered peers because they are all equal and share resources among each other without requiring a server. Since the workgroup doesn't share a common security and resource database, users and resources must be defined on each computer. Joining a workgroup requires all participants to use a matching name; all Windows computers (Windows 7, 8 and 10) are automatically assigned to a default group

user must know the name of the workgroup that computer belongs to plus the username and password of an account on the remote computer.

The main disadvantages of workgroups are:

- If a user account will be used for accessing resources on multiple machines, the user account will need to be created on those machines this requires that the same username and password be used.
- The low security protocol used for authentication between nodes.
- Desktop computers have a fixed limit of 15 or less connections. Note that this is in reference to connections to an individual desktop.

One of the most common mistakes when setting up a peer-to-peer network is misspelling the workgroup name on one of the computers. For example, suppose you decide that all the computers should belong to a workgroup named MYGROUP. If you accidentally spell the workgroup name MYGRUOP for one of the computers, that computer will be isolated in its own workgroup. If you can't locate a computer on your network, the workgroup name is one of the first things to check.

Windows Workgroups vs Homegroups and Domains

Domain Controller

Windows domains support client-server local networks. A specially configured computer called **Domain Controller** running a Windows Server operating system serves as a central server for all clients. Windows domains can handle more computers than workgroups due to the ability to maintain centralized resource sharing and access control. A client PC can belong to either to a workgroup or to a Windows domain, but not both. Assigning a computer to the domain automatically removes it from the workgroup.

Microsoft HomeGroup

Microsoft introduced the Homegroup concepts in Windows 7. Homegroups are designed to simplify the management of workgroups for administrators,

particularly homeowners. Instead of requiring an administrator to manually set up shared user accounts on every PC, HomeGroup security settings can be managed through one shared login.

Joining a Homegroup does not remove a PC from its Windows WORKGROUP, the two sharing methods co-exist. Computers running versions of Windows operating systems older than Windows 7 (like XP and vista), however cannot be members of HomeGroups.

Domain Controllers

A domain controller (DC) is a server computer that responds to security authentication requests within a computer network domain. It is a network server that is responsible for allowing end devices to access shared domain resources. It authenticates users, stores user account information and enforces security policy for a domain. It is most commonly implemented in Microsoft Windows environments (see below about Windows Domain), where it is the centerpiece of the Windows Active Directory service. However, non-Windows domain controllers can be established via identity management software such as Samba (see the last paragraph of section 2.1).

Domain controllers are typically deployed as a cluster to ensure high-availability and maximize reliability. In a Windows environment, one domain controller serves as the Primary Domain Controller (PDC) and all other servers promoted to domain controller status in the domain server as a Backup Domain Controller (BDC). In Unix-based environments, one machine serves as the master domain controller and others serve as replica domain controllers, periodically replicating database information from the main domain controller and storing it in a read-only format.

On Microsoft Servers, a domain controller (DC) is a server computer that responds to security authentication requests (logging in, etc.) within a Windows domain. A Windows domain is a form of a computer network in which all user accounts, computers, printers and other security principals, are registered with a central database located on one or more clusters of central computers known as domain controllers. A domain is a concept introduced in Windows NT whereby a user may be granted access to a number of computer resources with the use of a single username and password combination. You must setup at least one Domain Controller in every Windows domain..

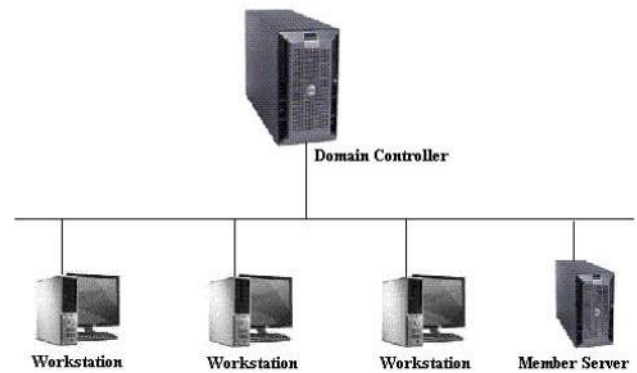


Figure. Domain Controller

Windows Server can be one of three kinds: Active Directory “domain controllers” (ones that provide identity and authentication), Active Directory “member servers” (ones that provide complementary services such as file repositories and schema) and Windows Workgroup “stand-alone servers”. The term “Active Directory Server” is sometimes used by Microsoft as synonymous to “Domain Controller” but the term is discouraged.

System requirements for a Domain Controller

This section outlines the minimum hardware requirements to run the latest Windows Server available as this resource is prepared (i.e. Windows Server 2022). If your computer has less than the minimum requirements, you will not be able to install the server correctly. Actual requirements will vary based on your system configuration and the applications and features you install.

Processor

Processor performance depends not only on the clock frequency of the processor, but also on the number of processor cores and the size of the processor cache. The following are the minimum processor requirements for the product:

- 1.4 GHz 64-bit processor

RAM

The following are the estimated minimum RAM requirements for the product:

- 512 MB (2 GB for Server with Desktop

Experience installation option)

Storage controller and disk space requirements
Computers that run Windows Server must include a storage adapter that is compliant with the PCI Express architecture specification. Persistent storage devices on servers classified as hard disk drives must not be PATA. Windows Server does not allow ATA/PATA/IDE/EIDE for boot, page, or data drives. The estimated **minimum** disk space requirements for the system partition is 32 GB

- Add a new entry
- Delete an entry

Network adapter requirements

Network adapters used with this latest release should include an Ethernet adapter capable of at least **1 gigabit** per second throughput.

LDAP & Windows Active Directory

The Lightweight Directory Access Protocol (LDAP) is an open, vendor-neutral, industry standard application protocol for accessing and maintaining distributed directory information services over an Internet Protocol (IP) network. Directory services play an important role in developing intranet and Internet applications by allowing the sharing of information about users, systems, networks, services, and applications throughout the network. As examples, directory services may provide any organized set of records, often with a hierarchical structure, such as a corporate email directory. Similarly, a telephone directory is a list of subscribers with an address and a phone number.

Protocol overview

A client starts an LDAP session by connecting to an LDAP server, called a Directory System Agent (DSA), by default on TCP and UDP port 389, or on port 636 for LDAPS (LDAP over TLS/SSL, see below). The client then sends an operation request to the server, and a server sends responses in return. With some exceptions, the client does not need to wait for a response before sending the next request, and the server may send the responses in any order. All information is transmitted using Basic Encoding Rules (BER).

The client may request the following operations:

- StartTLS– use LDAPv3 Transport Layer Security (TLS) extension for a secure connection
- Bind – authenticate and specify LDAP protocol version
- Search – search for and/or retrieve directory entries
- Compare – test if a named entry contains a given attribute value

- Modify an entry
- Modify Distinguished Name (DN) – move or rename an entry
- Abandon – abort a previous request
- Extended Operation – generic operation used to define other operations
- Unbind – close the connection (not the inverse of Bind)

A common alternative method of securing LDAP communication is using an SSL tunnel. The default port for LDAP over SSL is 636. The use of LDAP over SSL was common in LDAP Version 2 (LDAPv2) but it was never standardized in any formal specification. This usage has been deprecated along with LDAPv2, which was officially retired in 2003.

- Domain Services (DS)
- Lightweight Directory Services (LDS)
- Certificate Services (CS)
- Federation Services (FS)
- Rights Management Services (RMS)

Windows Active Directory

Active Directory (AD) is a directory service developed by Microsoft for Windows domain networks. It is included in most Windows Server operating systems as a set of processes and services. Initially, it was used only for centralized domain management. However, it eventually became an umbrella title for a broad range of directory-based identity-related services.

A server running the Active Directory Domain Service (AD DS) role is called a domain controller. It authenticates and authorizes all users and computers in a Windows domain type network, assigning and enforcing security policies for all computers, and installing or updating software. For example, when a user logs into a computer that is part of a Windows domain, Active Directory checks the submitted password and determines whether the user is a system administrator or normal user. Also, it allows management and storage of information, provides authentication and authorization mechanisms, and establishes a framework to deploy other related services: Certificate Services, AD Federation Services, Lightweight Directory Services, and Rights Management Services. Active Directory uses LDAP versions 2 and 3, Microsoft's version of Kerberos, and DNS.

Active Directory Services

Active Directory Services consist of multiple directory services. The best known is Active Directory Domain Services, commonly abbreviated as AD DS or simply AD.

AD Logical Structure

As a directory service, an Active Directory instance consists of a database and corresponding executable code responsible for servicing requests and maintaining the database.

Objects

Active Directory structures are arrangements of information about objects. The objects fall into two broad categories: resources (e.g., printers) and security principals (user or computer accounts and groups). Security principals are assigned unique security identifiers (SIDs).

Each object represents a single entity—whether a user, a computer, a printer, or a group—and its attributes. Certain objects can contain other objects. An object is uniquely identified by its name and has a set of attributes—the characteristics and information that the object represents—defined by a schema, which also determines the kinds of objects that can be stored in Active Directory.

The schema object lets administrators extend or modify the schema when necessary. However, because each schema object is integral to the definition of Active Directory objects, deactivating or changing these objects can fundamentally change or disrupt a deployment. Schema changes automatically propagate throughout the system. Once created, an object can only be deactivated—not deleted. Changing the schema usually requires planning.

Forests, trees, and domains

The Active Directory framework that holds the objects can be viewed at a number of levels. The forest, tree, and domain are the logical divisions in an Active Directory network.

Within a deployment, objects are grouped into domains. The objects for a single domain are stored in a single database (which can be replicated). **Domains** are identified by their DNS name structure, the namespace.

A **domain** is defined as a logical group of network objects (computers, users, devices) that share the same Active Directory database.

A **tree** is a collection of one or more domains and domain trees in a contiguous namespace, and is linked in a transitive trust hierarchy.

At the top of the structure is the **forest**. A **forest** is a collection of trees that share a common global catalog, directory schema, logical structure, and directory

within which users, computers, groups, and other objects are accessible.

Organizational Units

The objects held within a domain can be grouped into organizational units (OUs). OUs can provide hierarchy to a domain, ease its administration, and can resemble the organization's structure in managerial or geographical terms. Microsoft recommends using OUs rather than domains for structure and to simplify the implementation of policies and administration. The OU is the recommended level at which to apply group policies, which are Active Directory objects formally named group policy objects (GPOs), although policies can also be applied to domains or sites (see below). The OU is the level at which administrative powers are commonly delegated, but delegation can be performed on individual objects or attributes as well.

AD Physical Structure

Sites are physical (rather than logical) groupings defined by one or more IP subnets. AD also holds the definitions of connections, distinguishing low-speed (e.g., WAN, VPN) from high-speed (e.g., LAN) links. Site definitions are independent of the domain and OU structure and are common across the forest. Sites are used to control network traffic generated by replication and also to refer clients to the nearest domain controllers (DCs).

Replication

Active Directory synchronizes changes using multi-master replication. Replication by default is 'pull' rather than 'push', meaning that replicas pull changes from the server where the change was effected. The Knowledge Consistency Checker (KCC) creates a replication topology of site links using the defined sites to manage traffic. Intra-site replication is frequent and automatic as a result of change notification, which triggers peers to begin a pull replication cycle. Inter-site replication intervals are typically less frequent and does not use change notification by default, although this is configurable and can be made identical to intra-site replication. Replication of Active Directory uses Remote Procedure Calls (RPC) over IP (RPC/IP).

AD Implementation

In general, a network utilizing Active Directory has

more than one licensed Windows server computer. Backup and restore of Active Directory is possible for a network with a single domain controller, but Microsoft recommends more than one domain controller to provide automatic failover protection of the directory. Domain controllers

are also ideally single-purpose for directory operations only, and should not run any other software or role.

Certain Microsoft products such as SQL Server and Exchange can interfere with the operation of a domain controller, necessitating isolation of these products on additional Windows servers. Combining them can make configuration or troubleshooting of either the domain controller or the other installed software more difficult. A business intending to implement Active Directory is therefore recommended to purchase a number of Windows server licenses, to provide for at least two separate domain controllers, and optionally, additional domain controllers for performance or redundancy, a separate file server, a separate Exchange server, a separate SQL Server, and so forth to support the various server roles.

Physical hardware costs for the many separate servers can be reduced through the use of virtualization, although for proper failover protection, Microsoft recommends not running multiple virtualized domain controllers on the same physical hardware.

AD Trusting

To allow users in one domain to access resources in another, Active Directory uses trusts. Trusts inside a forest are automatically created when domains are created. The forest sets the default boundaries of trust, and implicit, transitive trust is automatic for all domains within a forest.

Terminology

One-way trust: One domain allows access to users on another domain, but the other domain does not allow access to users on the first domain.

Two-way trust: Two domains allow access to users on both domains.

Trusted domain: The domain that is trusted; whose users have access to the trusting domain.

Transitive trust: A trust that can extend beyond two domains to other trusted domains in the forest.

Intransitive trust: A one way trust that does not extend beyond two domains.

Explicit trust: A trust that an admin creates. It is not transitive and is one way only.

Cross-link trust: An explicit trust between domains in different trees or in the same tree when a descendant/ancestor (child/parent) relationship does not exist between the two domains.

Shortcut: Joins two domains in different trees, transitive, one- or two-way.

Forest trust: Applies to the entire forest. Transitive, one- or two-way.

Realm: Can be transitive or nontransitive (intransitive), one- or two-way.

External: Connect to other forests or non-AD domains. Nontransitive, one- or two-way.

PAM trust: A one-way trust used by Microsoft Identity Manager from a (possibly low-level) production forest to a (Windows Server 2016 functionality level) 'bastion' forest, which issues time-limited group memberships.

AD Management solutions

Microsoft Active Directory management tools include:

- Active Directory Administrative Center (Introduced with Windows Server 2012 and above),
- Active Directory Users and Computers,
- Active Directory Domains and Trusts,
- Active Directory Sites and Services,
- ADSI Edit,
- Local Users and Groups,
- Active Directory Schema snap-ins for Microsoft Management Console (MMC),
- SysInternals ADEplorer

These management tools may not provide enough functionality for efficient workflow in large environments. Some third-party solutions extend the administration and management capabilities. They provide essential features for a more convenient administration processes, such as automation, reports, integration with other services, etc.

Review Questions

1. Discuss the difference between Workgroup and Homegroup.
2. What are the system requirements of domain controller?
3. Discuss some of the active directory services.
4. To allow one user from one domain to use services in other domain, active directory uses trust.
5. Discuss the different terminologies used in trusting.
6. Discuss the difference between forests, trees and domains.
7. Discuss the logical and physical structure of domains.

Users and capabilities

A user account is a collection of settings and information that tells Windows which files and folders you can access,

CHAPTER THREE: USER ADMINISTRATION CONCEPTS & MECHANISMS

what you can do on your computer, what are your preferences, and what network resources you can access when connected to a network.

The user account allows you to authenticate to Windows or any other operating system so that you are granted authorization to use them. Multi-user operating systems such as Windows don't allow a user to use them without having a user account.

A user account in Windows is characterized by the following attributes:

User name: the name you are giving to that account.

Password: the password associated with the user account (in Windows 7 or older versions you can also use blank passwords).

User group: a collection of user accounts that share the same security rights and permissions. A user account must be a member of at least one user group.

Type: all user accounts have a type which defines their permissions and what they can do in Windows.

Administrator: The "Administrator" user account has complete control over the PC. He or she can install anything and make changes that affect all users of that PC.

Standard: The "Standard" user account can only use the software that's already installed by the administrator and change system settings that don't affect other users.

Guest: The "Guest" account is a special type of user account that has the name Guest and no password. This is only for users that need temporary access to the PC. This user can only use the software that's already installed by the administrator and cannot make any changes to system settings.

All user accounts have specific capabilities, privileges, and rights. When you create a user account, you can grant the user specific capabilities by making the user a member of one or more groups. This gives the user the capabilities of these groups. You then assign additional capabilities by making a user a member of the appropriate groups or withdraw capabilities by removing a user from a group.

An important part of an administrator's job is being able to determine and set permissions, privileges, and logon rights as necessary. Although you can't change a group's built-in capabilities, you can change a group's default privileges and logon rights. For example, you could

revoke network access to a computer by removing a group's right to access the computer from the network.

What is File & Folder Permissions?

Permissions are a method for assigning access rights to specific user accounts and user groups. Through the use of permissions, Windows defines which user accounts and user groups can access which files and folders, and what they can do with them. To put it simply, permissions are the operating system's way of telling you what you can or cannot do with a file or folder.

On Windows operating system, to learn the permissions of any folder, right click on it and select "Properties." In the Properties window, go to the Security tab. In the "Group or user names" section you will see all the user accounts and user groups that have permissions to that folder. If you select a group or a user account, then see its assigned permissions, in the "Permissions for Users" section.

In Windows, a user account or a user group can receive one of the following permissions to any file or folder:

Read: allows the viewing and listing of a file or folder. When viewing a folder, you can view all its files and subfolders.

Write: allows writing to a file or adding files and subfolders to a folder.

List folder contents: this permission can be assigned only to folders. It permits the viewing and listing of files and subfolders, as well as executing files that are found in that folder.

Read & execute: permits the reading and accessing of a file's contents as well as its execution. When dealing with folders, it allows the viewing and listing of files and subfolders, as well as the execution of files.

Modify: when dealing with files, it allows their reading, writing and deletion. When dealing with folders, it allows the reading and writing of files and subfolders, plus the deletion of the folder.

Full control: it allows reading, writing, changing and deleting of any file and subfolder. Generally, files inherit the permissions of the folder where they are placed, but users can also define specific permissions that are assigned only to a specific file. To make your computing life simpler, it is best to edit permissions only at a folder level.

The most efficient way to assign user rights is to make the user a member of a group that already has the right. In some cases, however, you might want a user to have a particular right but not have all the other rights of the group. One way to resolve this problem is to give the user

Assigning User Rights

the rights directly. Another way to resolve this is to create a special group for users that need the right. This is the approach used with the Remote Desktop Users group, which was created by Microsoft to grant Allow Logon through Terminal Services to groups of users.

You assign user rights through the Local Policies node of Group Policy. Local policies can be set on a per-computer basis using a computer's local security policy or on a domain or OU basis through an existing group policy for the related domain or OU. When you do this, the local policies apply to all accounts in the domain or OU.

What is Roaming profile?

A Windows profile is a set of files that contains all settings of a user including per-user configuration files and registry settings. In an Active Directory or NT4 domain you can set that the profile of a user is stored on a server. This enables the user to log on to different Windows domain members and use the same settings.

When using roaming user profiles, a copy of the profile is downloaded from the server to the Windows domain member when a user logs into. Until the user logs out, all settings are stored and updated in the local copy. During the log out, the profile is uploaded to the server.

Assigning a Roaming Profile to a User

Depending on the Windows version, Windows uses different folders to store the roaming profile of a user. However, when you set the profile path for a user, you always set the path to the folder without any version suffix. For example:

\\server\profiles\user_name

A roaming user profile is a file synchronization concept in the Windows NT family of operating systems that allows users with a computer joined to a Windows domain to log on to any computer on the same domain and access their documents and have a consistent desktop experience, such as applications remembering toolbar positions and preferences, or the desktop appearance staying the same, while keeping all related files stored locally, to not continuously depend on a fast and reliable network connection to a file server.

All Windows operating systems since Windows NT 3.1 are designed to support roaming profiles. Normally, a standalone computer stores the user's documents, desktop items, application preferences, and desktop appearance on the local computer in two divided sections, consisting of the portion that could roam plus an additional temporary portion containing items such as the web browser cache.

The Windows Registry is similarly divided to support roaming; there are System and Local Machine hives that

stay on the local computer, plus a separate User hive (HKEY CURRENT USER) designed to be able to roam with the user profile.

When a roaming user is created, the user's profile information is instead stored on a centralized file server accessible from any network-joined desktop computer. The login prompt on the local computer checks to see if the user exists in the domain rather than on the local computer; no preexisting account is required on the local computer. If the domain login is successful, the roaming profile is copied from the central file server to the desktop computer, and a local account is created for the user.

When the user logs off from the desktop computer, the user's roaming profile is merged from the local computer back to the central file server, not including the temporary local profile items. Because this is a merge and not a move/delete, the user's profile information remains on the local computer in addition to being merged to the network.

When the user logs in on a second desktop computer, this process repeats, merging the roaming profile from the server to the second desktop computer, and then merging back from the desktop to the server when the user logs off.

When the user returns to the first desktop computer and logs in, the roaming profile is merged with the previous profile information, replacing it. If profile caching is enabled, the server is capable of merging only the newest files to the local computer, reusing the existing local files that have not changed since the last login, and thereby speeding up the login process.

Windows stores information about a particular user in a so-called profile. Some examples of the sort of data that gets stored in a profile are (N.B. this list is not exhaustive):

Application data and settings

The "Documents"/"My Documents" folder

The "Downloads" folder, which is where your internet browser may save to by default
Files stored on your Desktop

Directories you create under c:\users\[your-username]

Members of some groups in the department have a roaming profile. This means that the master copy of the profile is stored on a fileserver. When you log in to a Windows computer, the contents of your profile will be

synchronized from the fileserver to the local computer. When you log out of the computer, any changes to the profile are then synchronized back to the server. Instructions for checking whether or not you have a roaming profile are available.

stores low-level settings for the Microsoft Windows operating system and for applications that opt to use the

There are two main reasons why a roaming profile might be useful in the department. Firstly, because the contents of the profile are stored centrally, whenever you log on to any computer in the department you will have the same application data and settings (e.g., internet browser bookmarks, preferences in Microsoft Office etc.).

Secondly, because the master copy of your roaming profile is stored on a Departmentally-managed fileserver, all data stored within it is automatically backed up.

What are the main differences of roaming and local profiles?

Windows roaming and local profiles are similar in that they both store Windows user settings and data. A local profile is one that is stored directly on the computer. The main advantage to using a local profile is that the profile is accessible even when the computer is disconnected from the network. A major drawback of a local profile is that the user profile data is not being automatically backed up by the server. Since most users rarely back up their computers, if a hard drive fails, any data that is stored within local profiles on that machine would be lost.

Roaming profiles are stored on a server and can be accessed by logging into any computer on the network. In a roaming profile, when a user logs onto the network, his/her profile is copied from the server to the user's desktop. When the user logs off of their computer, the profile (including any changes that the user might have made) is copied back to the server. A major drawback of roaming profiles is that they can slow down the network. Windows user profiles often become very large as the user profile data continues to grow. If you have a large roaming profile, the login and logoff times may take a significant amount of time.

The solution to this problem is to use folder redirection with roaming profiles. Folder redirection allows specific folders (such as the Desktop and Documents folder) to be permanently stored on the server. Doing so eliminates the need for the redirected folder to be copied as a part of the login and logoff processes.

Advanced Concepts I

The Registry

The Windows Registry is a hierarchical database that

registry. The kernel, device drivers, services, Security Accounts Manager, and user interface can all use the registry. The registry also allows access to counters for profiling system performance.

Registry Structure

Keys and values

The registry contains two basic elements: keys and values. Registry keys are container objects similar to folders. Registry values are non-container objects similar to files. Keys may contain values and subkeys. Keys are referenced with a syntax similar to Windows' path names, using backslashes to indicate levels of hierarchy. Keys must have a case insensitive name without backslashes.

The hierarchy of registry keys can only be accessed from a known root key handle (which is anonymous but whose effective value is a constant numeric handle) that is mapped to the content of a registry key preloaded by the kernel from a stored "hive", or to the content of a subkey within another root key, or mapped to a registered service or DLL that provides access to its contained subkeys and values.

Text

Box:

Example:

HKEY_LOCAL_MACHINE\Software\Microsoft\Windows refers to the subkey "Windows" of the subkey "Microsoft" of the subkey "Software" of the HKEY_LOCAL_MACHINE root key.

There are seven predefined root keys, traditionally named according to their constant handles defined in the Win32 API, or by synonymous abbreviations (depending on applications):

HKEY_LOCAL_MACHINE or

HKLM

HKEY_CURRENT_CONFIG or

HKCCHKEY_CLASSES_ROOT or

HKCR HKEY_CURRENT_USER

or HKCU HKEY_USERS or HKU

HKEY_PERFORMANCE_DATA (only in

WindowsNT, but invisible in the Windows Registry

Editor) HKEY_DYN_DATA (only in Windows 9x,

and visible in the Windows Registry Editor)

Like other files and services in Windows, all registry keys may be restricted by access control lists (ACLs), depending on user privileges, or on security tokens acquired by applications, or on system

security policies enforced by the system (these restrictions may be predefined by the system itself, and configured by local system administrators or by domain administrators). Different users, programs, services or remote systems may only see some parts of the hierarchy or distinct hierarchies from the same root keys.

depending on your use case, needs, or company policy.

Advanced Concepts II

Routing and NAT

Routing refers to establishing the routes that data packets take on their way to a particular destination. This term can be applied to data traveling on the Internet, over 3G or 4G networks, or over similar networks used for telecom and other digital communications setups. Routing can also take place within proprietary networks.

In general, routing involves the network topology, or the setup of hardware, that can effectively relay data. Standard protocols help to identify the best routes for data and to ensure quality transmission. Individual pieces of hardware such as routers are referred to as “nodes” in the network. Different algorithms and protocols can be used to figure out how to best route data packets, and which nodes should be used. For example, some data packets travel according to a distance vector model that primarily uses distance as a factor, whereas others use Link-State Protocol, which involves other aspects of a “best path” for data.

Data packets are also made to give networks information. Headers on packets provide details about origin and destination. Standards for data packets allow for conventional design, which can help with future routing methodologies. As the world of digital technology evolves, routing will also evolve according to the needs and utility of a particular network.

Network Address Translation (NAT)

NAT translates the IP addresses of computers in a local network to a single IP address. This address is often used by the router that connects the computers to the Internet. The router can be connected to a DSL modem, cable modem, T1 line, or even a dial-up modem. When other computers on the Internet attempt to access computers within the local network, they only see the IP address of the router. This adds an extra level of security, since the router can be configured as a firewall, only allowing authorized systems to access the computers within the network.

Proxies and Gateways

What is proxy server?

A proxy server acts as a gateway between you and the Internet. It’s an intermediary server separating end users from the websites they browse. Proxy servers provide varying levels of functionality, security, and privacy

Here's a few of the primary uses for a proxy server:

- Firewalls
- Content filters
- Bypassing content filters
- Caching
- Security
- Sharing Internet connections

- IoT gateways
- Email security gateways
- VoIP trunk gateways

What is a Gateway ?

A gateway is a node (router) in a computer network, a key stopping point for data on its way to or from other networks. Thanks to gateways, we are able to communicate and send data back and forth. The Internet wouldn't be any use to us without gateways (as well as a lot of other hardware and software).

How gateways work?

All networks have a boundary that limits communication to devices that are directly connected to it. Due to this, if a network wants to communicate with devices, nodes or networks outside of that boundary, they require the functionality of a gateway. A gateway is often characterized as being the combination of a router and a modem.

The gateway is implemented at the edge of a network and manages all data that is directed internally or externally from that network. When one network wants to communicate with another, the data packet is passed to the gateway and then routed to the destination through the most efficient path. In addition to routing data, a gateway will also store information about the host network's internal paths and the paths of any additional networks that are encountered.

Gateways are basically protocol converters, facilitating compatibility between two protocols and operating on any layer of the open systems interconnection (OSI) model.

Types of gateways

Gateways can take several forms and perform a variety of tasks. Examples of this include:

- Web application firewalls
- Cloud storage gateways

entire system) fail to run at all. In either case, performance

Review Questions

1. Discuss the different types of users and types of permissions they may have on the system.
2. Discuss the seven predefined root key in Windows registry.
3. What is routing and routing protocols? Discuss some of well-known routing protocols.
4. What is Network Address Translation (NAT)? Why do we need NAT in our LAN?
5. What are the benefits of NAT to an organization? Discuss the different ways used to implement NAT.
6. Discuss the meaning, benefit and how proxies work.
7. Discuss the different types of gateways.

CHAPTER 4: RESOURCE MONITORING & MANAGEMENT

Resource Monitoring & Management

As stated earlier, a great deal of system administration revolves around resources and their efficient use. By balancing various resources against the people and programs that use those resources, you waste less money and make your users as happy as possible. However, this leaves two questions:

What are resources?

How it is possible to know what resources are being used (and to what extent)?

The purpose of this chapter is to enable you to answer these questions by helping you to learn more about resources and how they can be monitored.

Before you can monitor resources, you first have to know what resources there are to monitor. All systems have the following resources available:

- CPU power
- Bandwidth
- Memory
- Storage

System Performance Monitoring

System performance monitoring is normally done in response to a performance problem. Either the system is running too slowly, or programs (and sometimes even the

monitoring is normally done as the first and last steps of a three-step process:

Monitoring to identify the nature and scope of the resource shortages that are causing the performance problems.

The data produced from monitoring is analyzed and a course of action (normally performance tuning and/or the procurement of additional hardware) is taken to resolve the problem

Monitoring System Capacity

Monitoring system capacity is done as part of an ongoing capacity planning program. Capacity planning uses long-term resource monitoring to determine rates of change in the utilization of system resources. Once these rates of change are known, it becomes possible to conduct more accurate long-term planning regarding the procurement of additional resources.

What to Monitor?

As stated earlier, the resources present in every system are CPU power, bandwidth, memory, and storage. At first glance, it would seem that monitoring would need only consist of examining these four different things.

Unfortunately, it is not that simple. For example, consider a disk drive. What things might you want to know about its performance?

- How much free space is available?
- How many I/O operations on average does it perform each second?
- How long on average does it take each I/O operation to be completed?
- How many of those I/O operations are reads? How many are writes?
- What is the average amount of data read/written with each I/O?

Monitoring CPU Power

In its most basic form, monitoring CPU power can be no more difficult than determining if CPU

utilization ever reaches 100%. If CPU utilization stays below 100%, no matter what the system is doing, there is additional processing power available for more work.

Monitoring Storage

However, it is a rare system that does not reach 100% CPU utilization at least some of the time. At that point it is important to examine more detailed CPU utilization data. By doing so, it becomes possible to start determining where the majority of your processing power is being consumed.

Monitoring Bandwidth

Monitoring bandwidth is more difficult than the other resources described here. The reason for this is due to the fact that performance statistics tend to be device-based, while most of the places where bandwidth is important tend to be the buses that connect devices. In those instances where more than one device shares a common bus, you might see reasonable statistics for each device, but the aggregate load those devices place on the bus would be much greater.

Some of the more common bandwidth-related statistics are:

- Bytes received/sent
- Interface counts and rates
- Transfers per Second

Monitoring Memory

If there is one area where a wealth of performance statistics can be found, it is in the area of monitoring memory utilization. Due to the inherent complexity of today's demand-paged virtual memory operating systems, memory utilization statistics are many and varied. It is here that the majority of a system administrator's work with resource management takes place.

The following statistics represent a cursory overview of commonly-found memory management statistics:

- Page Ins/Page Outs
- Active/Inactive Pages
- Free, Shared, Buffered, and Cached Pages
- Swap Ins/Swap Outs

Successfully monitoring memory utilization requires a good understanding of how demand-paged virtual memory operating systems work, which alone could take up an entire book.

Monitoring storage normally takes place at two different levels:

- Monitoring for sufficient disk space
- Monitoring for storage-related performance problems

The reason for this is that it is possible to have dire problems in one area and no problems whatsoever in the other. For example, it is possible to cause a disk drive to run out of disk space without once causing any kind of performance-related problems. Likewise, it is possible to have a disk drive that has 99% free space, yet is being pushed past its limits in terms of performance.

In any case, the following statistics are useful for monitoring storage:

- Free Space
- File System-Related Statistics
- Transfers per Second
- Reads/Writes per Second

Monitoring Tools

As your organization grows, so does the number of servers, devices, and services you depend on. The term **system** covers all of the computing resources of your organization. Each element in the system infrastructure relies on underlying services or provides services to components that are closer to user.

In networking, it is typical to think of a system as a **layered stack**. User software sits at the top of the stack and system applications and services on the next layer down. Beneath the services and applications, you will encounter operating systems and firmware. The performance of software elements needs to be monitored as an application stack.

Users will notice performance problems with the software that they use, but those problems rarely arise within that software. All layers of the application stack need to be examined to find the root cause of performance issues. You need to head off problems with real-time status monitoring before

they occur. **Monitoring tools** help you spot errors and service failures before they start to impact users.

Why do System Performance Monitoring?

Knowing whether a computer has issues is fairly straightforward when the computer is right in front of you. ***Knowing what's causing the problem? That's harder.*** But a computer sitting by itself is not as useful as it could be. Even the smallest small-office/home-office network has multiple nodes: laptops, desktops, tablets, WiFi access points, Internet gateway, smartphones, file servers and/or media servers, printers, and so on. That means you are in charge of “infrastructure” rather than just “equipment.” Any component might start misbehaving and could cause issues for the others.

Bandwidth monitoring tools and NetFlow and sFlow based traffic analyzers help you stay aware of the activity, capacity, and health of your network. They allow you to watch traffic as it flows through routers and switches, or arrive at and leaves hosts.

System Monitoring Software Essentials

In order to keep your system fit for purpose, your monitoring activities need to cover the following priorities:

- Acceptable delivery speeds
- Constant availability
- Preventative maintenance
- Software version monitoring and patching
- Intrusion detection
- Data integrity
- Security monitoring
- Attack mitigation
- Virus prevention and detection

Lack of funding may cause you to compromise on monitoring completeness. The expense of monitoring can be justified because of it:

- reduces user/customer support costs
- prevents loss of income caused by system outages or attack vulnerability
- prevents data leakage leading to litigation
- prevents hardware damage and loss of business-critical data

Minimum system monitoring software capabilities

- More sophisticated system monitoring package provides a much broader range of capabilities, such as:
- Monitoring multiple servers. Handling servers from various vendors running various operating systems. Monitoring servers at multiple sites and in cloud environments.
- Monitoring a range of server metrics: availability, CPU usage, memory usage, disk space, response time, and upload/download rates. Monitoring CPU temperature and power supply voltages.
- Monitoring applications. Using deep knowledge of common applications and services to monitor key server processes, including web servers, database servers, and application stacks.
- Automatically alerting you of problems, such as servers or network devices that are overloaded or down, or worrisome trends. Customized alerts that can use multiple methods to contact you – email, SMS text messages, pager, etc.
- Triggering actions in response to alerts, to handle certain classes of problems automatically.
- Collecting historical data about server and device health and behavior.
- Displaying data. Crunching the data and analyzing trends to display illuminating visualizations of the data.
- Reports. Besides displays, generating useful predefined reports that help with tasks like forecasting capacity, optimizing resource usage, and predicting needs for maintenance and upgrades.
- Customizable reporting. A facility to help you create custom reports.
- Easy configurability, using methods like auto-discovery and knowledge of server and application types.
- Non-intrusive: imposing a low overhead on your production machines and services. Making smart use of agents to offload monitoring where appropriate.
- Scalability: Able to grow with your business, from a small or medium business (SMB) to a large enterprise.

Windows Task Manager

Task Manager (old name Windows Task Manager) is a task manager, system monitor, and startup manager included with all versions of Microsoft Windows since Windows NT 4.0 and Windows 2000.

- Performance
- App History

Windows Task Manager provides information about computer performance and shows detailed information about the programs and processes running on the computer, including name of running processes, CPU load, commit charge, I/O details, logged-in users, and Windows services; if connected to the network, you can also view the network status and quickly understand how the network works.

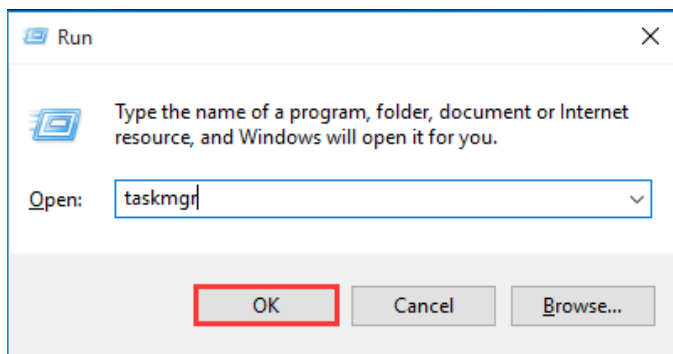
Microsoft improves the task manager between each version of Windows, sometimes quite dramatically. Specifically, the task managers in Windows 10 and Windows 8 are very different from those in Windows 7 and Windows Vista, and the task managers in Windows 7 and Vista are very different from those in Windows XP. A similar program called Tasks exists in Windows 98 and Windows 95.

How to Open the Task Manager?

Starting **Task Manager** is always a concern for many of you. Now we will list some easy and quick ways for you to open it. Some of them might come in handy if you don't know how to open a Task Manager or you can't open Task Manager the way you're used to.

If you prefer using a mouse over a keyboard, one of the quickest ways to launch **Task Manager** is to **right-click** on any blank area on the **taskbar** and select Task Manager. Just need two clicks.

You can also run Task Management by hitting **Windows** to open the Run box, typing **taskmgr** and then hitting Enter or clicking OK.



- Processes

- Startup
- Users
- Details
- Services

Windows Resource Monitoring

Resource Monitor is a system application included in Windows Vista and later versions of Windows that allows users to look at the presence and allocation of resources on a computer. This application allows administrators and other users determine how system resources are being used by a particular hardware setup.

How to start Resource Monitor?

Users and administrators have several options to start Resource Monitor. It is included in several versions of Windows, and some options to start the tool are only available in select versions of the operating system.

The first two methods should work on all versions of Windows that are supported by Microsoft.

1. Windows-R to open the **run** box. Type resmon.exe, and hit the Enter-key.
2. Windows-R to open the **run** box. Type perfmon.exe /res, and hit the Enter-key.
3. On Windows 10: Start → All Apps → Windows Administrative Tools → Resource Monitor
4. Old Windows: Start → All Programs → Accessories → System Tools → Resource Monitor
5. Open **Task Manager** with **Ctrl+Shift+Esc** → Performance tab, click **open Resource Monitor**.

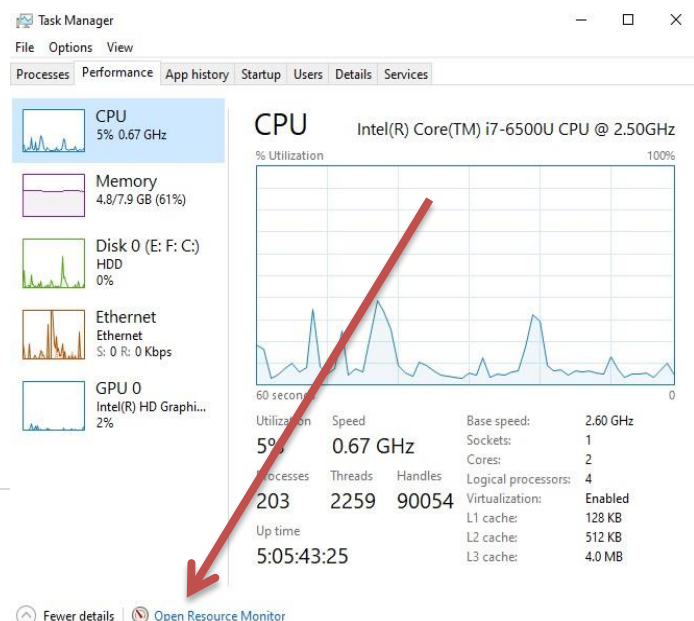


Figure: Opening Resource Monitor from Task Manager

The Resource Monitor interface looks the same on Windows 7, 8.1 and 10. The program uses tabs to separate data, it loads an overview when you start it, and including CPU, Memory, Disk, and Network are the five tabs of the program including all the processes that use the resources.

The sidebar displays graphs that highlight the CPU, Disk, Network, and Memory use over a period of 60 seconds.

Monitor CPU Usage with Resource Monitor

You need to switch to the CPU tab if you want to monitor CPU utilization in detail. You find the processes listing of the overview page there, and also the three new listings Services, Associated Handles and Associated Modules.

Memory in Resource Monitor

The memory tab lists processes just like the CPU tab does, but with a focus on memory usage. It features a physical memory view on top of that that visualizes the distribution of memory on the Windows machine.

If this is your first time accessing the information, you may be surprised that quite a bit of memory may be hardware reserved. The graphs highlight the used physical memory, the commit charge, and the hard faults per second. Each process is listed with its name and process ID, the hard faults, and various memory related information.

Commit: Amount of virtual memory reserved by the operating system for the process.

Working Set: Amount of physical memory currently in use by the process.

Shareable: Amount of physical memory in use by the process that can be shared with other processes.

Private: Amount of physical memory in use by the process that cannot be used by other processes.

Network Activity in Resource Monitor

The Network tab lists network activity, TCP connections and listening ports. It lists network activity of any running process in detail. It is useful, as it tells you right away if processes connect to the Internet.

Bandwidth

Bandwidth describes the maximum data transfer rate of a network. It measures how much data can be sent over a specific connection in a given amount of time. For example, a gigabit Ethernet connection has a bandwidth of 1,000 Mbps (125 megabytes per second). An Internet connection via cable modem may provide 25 Mbps of bandwidth.

The most common Internet bottleneck is your connection to your ISP.

Bandwidth vs. Speed

Internet speed is a major vice to any Internet user. Even though Internet speed and data transfer mostly revolve around bandwidth, your Internet speed can also be different from the Internet bandwidth expectations. What tends to make it complicated is that the terms bandwidth, speed, and bandwidth speed are used interchangeably, but they are actually different things. Most people refer to speed as how long it takes to upload and download files, videos, livestreams, and other content.

Network bandwidth

Use of bandwidth can also be monitored by a network bandwidth monitor. Network bandwidth is a fixed commodity. There are several ways to use network bandwidth. First, you can control the data flow in your Internet connection. That is you can streamline data from one point to another point. Next, you can also optimize data so that it consumes less bandwidth from what is allocated.

Network Printers

Network printing allows us to efficiently use printing resources. With network printing we first connect all of our work stations to a network and then we implement a network printer. In general there are two ways this can be done.

Remote Administration

Remote administration is an approach being followed to control either a computer system or a network or an application or all three from a remote location. Simply put, Remote administration refers to any method of controlling a computer from a remote location. A remote location may refer to a computer in the next room or one on the other side of the world. It may also refer to both legal and illegal remote administration. Generally, remote administration is essentially adopted when it is difficult or

impractical to a person to be physically present and do administration on a system's terminal.

compromise the system. It is advised that remote administration to be used only in emergency or essential

Requirements to Perform Remote Administration

- Internet connection
- Connecting

Common Tasks/Services for which Remote Administration is used

Generally, remote administration is needed for user management, file system management, software installation/configuration, network management, Network Security/Firewalls, VPN, Infrastructure Design, Network File Servers, Auto-mounting etc. and kernel optimization/ recompilation.

The following are some of the tasks/ services for which remote administration need to be done:

- Hacking
- Remote Desktop Solutions
- Controlling one's own computer from a remote location
- ICT Infrastructure Management

The following are some of proprietary and open source applications that can be used to achieve this.

- SSH (Secure Shell)
- OpenSSH (OpenBSD Secure Shell)
- Telnet
- rlogin
- rsh
- VNC (Virtual Network Computing)
- FreeNX
- Wireless Remote Administration
- Remote Desktop Connection (RDC)

Disadvantages of Remote Administration

Remote administration has many disadvantages too apart from its advantages. The first and foremost disadvantage is the security. Generally, certain ports to be open at Server level to do remote administration. Due to open ports, the hackers/attackers takes advantage to

situations only to do administration remotely. In normal situations, it is ideal to block the ports to avoid remote administration.

Performance

Redundant Array of Inexpensive (or Independent) Disks (RAID) RAID is a data storage virtualization technology that combines multiple physical disk drive components into one or more logical units for the purposes of data redundancy, performance improvement, or both. This was in contrast to the previous concept of highly reliable mainframe disk drives referred to as **Single Large Expensive Disk (SLED)**.

Data is distributed across the drives in one of several ways, referred to as **RAID levels**, depending on the required level of redundancy and performance. The different schemes, or data distribution layouts, are named by the word “RAID” followed by a number, for example RAID 0 or RAID 1. Each scheme, or RAID level, provides a different balance among the key goals: reliability, availability, performance, and capacity. RAID levels greater than RAID 0 provide protection against unrecoverable sector read errors, as well as against failures of whole physical drives.

Standard levels

Originally, there were five standard levels of RAID, but many variations have evolved, including several nested levels and many non-standard levels (mostly proprietary). RAID levels and their associated data formats are standardized by the **Storage Networking Industry Association (SNIA)** in the Common RAID **Disk Drive Format (DDF)** standard:

RAID 0 consists of striping, but no mirroring or parity. Compared to a spanned volume, the capacity of a RAID 0 volume is the same; it is the sum of the capacities of the drives in the set. But because striping distributes the contents of each file among all drives in the set, the failure of any drive causes the entire RAID 0 volume and all files to be lost. In comparison, a spanned volume preserves the files on the unfailing drives. The benefits of RAID 0 is that the throughput of read and write operations to any file is multiplied by the number of drives because, unlike spanned volumes, reads and writes are done concurrently.

RAID 1 consists of data mirroring, without parity or striping. Data is written identically to two or more drives, thereby producing a “mirrored set” of drives.

Thus, any read request can be serviced by any drive in the set. If a request is broadcast to every drive in the set, it can be serviced by the drive that accesses the data first (depending on its seek time and rotational latency),

improving performance. Sustained read throughput, if the controller or software is optimized for it, approaches the sum of throughputs of every drive in the set, just as for RAID 0.

Actual read throughput of most RAID 1 implementations is slower than the fastest drive. Write throughput is always slower because every drive must be updated, and the slowest drive limits the write performance. The array continues to operate as long as at least one drive is functioning.

RAID 2 consists of bit-level striping with dedicated Hamming-code parity. All disk spindle rotation is synchronized and data is striped such that each sequential bit is on a different drive. Hamming-code parity is calculated across corresponding bits and stored on at least one parity drive. This level is of historical significance only; as of 2014 it is not used by any commercially available system.

RAID 3 consists of byte-level striping with dedicated parity. All disk spindle rotation is synchronized and data is striped such that each sequential byte is on a different drive. Parity is calculated across corresponding bytes and stored on a dedicated parity drive. Although implementations exist, RAID 3 is not commonly used in practice.

RAID 4 consists of block-level striping with dedicated parity. The main advantage of RAID 4 over RAID 2 and 3 is I/O parallelism: in RAID 2 and 3, a single read I/O operation requires reading the whole group of data drives, while in RAID 4 one I/O read operation does not have to spread across all data drives. As a result, more I/O operations can be executed in parallel, improving the performance of small transfers.

RAID 5 consists of block-level striping with distributed parity. Unlike RAID 4, parity information is distributed among the drives, requiring all drives but one to be present to operate. Upon failure of a single drive, subsequent reads can be calculated from the distributed parity such that no data is lost. RAID 5 requires at least three disks. Like all single-parity concepts, large RAID 5 implementations are susceptible to system failures because of trends regarding array rebuild time and the chance of drive failure during rebuild. Rebuilding an array requires reading all data from all disks, opening a chance for a second drive failure and the loss of the entire array.

RAID 6 consists of block-level striping with double distributed parity. Double parity provides fault tolerance up to two failed drives. This makes larger RAID groups more practical, especially for high-availability systems, as large-capacity drives take longer to restore. RAID 6 requires a minimum of four disks. As with RAID 5, a single drive failure results in reduced performance of the

entire array until the failed drive has been replaced. With a RAID 6 array, using drives from multiple sources and

manufacturers, it is possible to mitigate most of the problems associated with RAID 5. The larger the drive capacities and the larger the array size, the more important it becomes to choose RAID 6 instead of RAID

5. RAID 10 also minimizes these problems.

- Redhat
- Kubuntu
- Debian

Review Questions

1. Discuss why we need resource monitoring in our infrastructure, and what are the resources that we are going to monitor.
2. Discuss the different kinds of resource monitoring tools that are already available in Windows operating systems.
3. Besides the free and already available resource monitoring and management tools mentioned above, discuss some of other well-known free and commercial tools available for system administrators.
4. Why remote administration is needed? Explain.
5. List the different network clients.
6. What are the different remote administration tools?

CHAPTER 5: SECURITY

Introduction

What is Unix/Linux?

The Unix OS is a set of programs that act as a link between the computer and the user. The computer program that allocates the system resources and coordinates all the details of the computer's internals is called the operating system or the kernel. Users communicate with the kernel through a program known as the shell. The shell is a command line interpreter; it translates commands entered by the user and converts them into a language that is understood by the kernel.

Linux Distribution

Linux distribution is an operating system that is made up of a collection of software based on Linux kernel or you can say distribution contains the Linux kernel and supporting libraries and software. Around 600+ Linux Distributions are available and some of the popular Linux distributions are:

- Linux Mint
- Ubuntu
- Solus
- Fedora
- OpenSUSE

Unix/Linux Architecture

implementations, you will find a lot of similarities in the different distributions. Linux may appear different

Here is a basic block diagram of a Unix system.

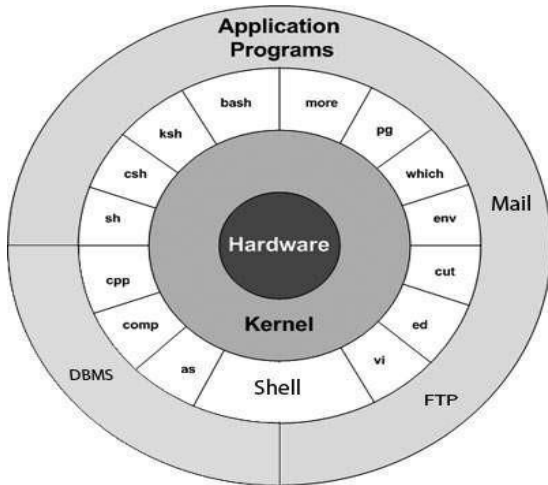


Figure. Block diagram of Unix system

The main concept that unites all the versions of Unix is the following four basics:

Kernel

Shell

Commands and Utilities

Properties of Linux

Linux Pros

A lot of the advantages of Linux are a consequence of Linux' origins deeply rooted in UNIX, except for the first advantage, of course:

- Linux is free
- Linux is portable to any hardware platform
- Linux was made to keep on running
- Linux is secure and versatile
- Linux is scalable
- The Linux OS and most Linux applications have very short debug-times

Linux Cons

- There are far too many different distributions
- Linux is not very user friendly and confusing for beginners

Linux and GNU

Although there are a large number of Linux

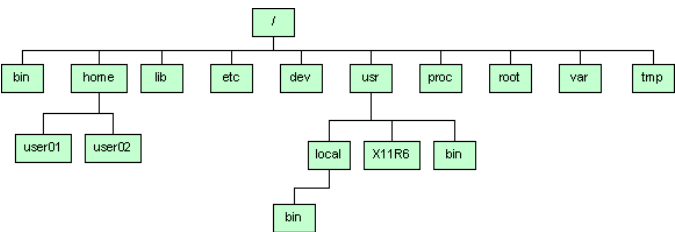
depending on the distribution, your hardware and personal taste, but the fundamentals on which all graphical and other interfaces are built, remain the same. The Linux system is based on GNU tools (Gnu's Not UNIX), which provide a set of standard ways to handle and use the system.

Sorts of Files

Most files are just files, called regular files; they contain normal data, for example text files, executable files or programs, input for or output from a program and so on. The -l option to ls displays the file type, using the first character of each input line:

The following table gives an overview of the characters determining the file type:

For convenience, the Linux file system is usually thought of in a tree structure as shown below:



The tree of the file system starts at the trunk or slash, indicated by a forward slash (/). This directory, containing all underlying directories and files, is also called root directory or “the root” of the file system.

Linux Directory Structure

/bin: Common programs, shared by the system, the system administrator and the users.

/boot: The startup files and the kernel, vmlinuz. In some recent distributions also grub data. Grub is the GRand Unified Boot loader and is an attempt to get rid of the many different boot-loaders we know today.

/dev: Contains references to all the CPU peripheral hardware, which are represented as files with special properties.

Symbol	Meaning
-	Regular file
d	Directory
l	Link
c	Special file
s	Socket
p	Named pipe
b	Block device

/etc: Most important system configuration files are in /etc, this directory contains data similar to those in the Control Panel in Windows.

/home Home directories of the common users.

/lib: Library files, includes files for all kinds of programs needed by the system and the users.

/lost+found: Every partition has a lost+found for files that were saved during failures are here.

/misc: For miscellaneous purposes.

/mnt: Standard mount point for external file systems, e.g. a CD-ROM or a digital camera.

/opt: Typically contains extra and third party software.

/proc: Virtual file system containing system resources information. You can type `man proc` command on terminal to see more information about the meaning of the files in `proc`.

/root: The administrative user's home directory. Mind the difference between `/`, the root directory and `/root`, the home directory of the root user.

/sbin: Programs for use by the system and the system administrator.

/tmp : Temporary space to be used by the system, and its contents will be cleaned upon reboot, so don't use this for saving any work!

/usr: Programs, libraries, documentation etc. for all user-related programs.

/var: Storage for all variable and temporary files created by users, such as log files, temporary files downloaded from the Internet, or to keep an image of a CD before burning it.

communications network. The protocols used by the machines are organized in different layers or levels, in

Linux Systems and Network Concepts

What is networking?

A network consists of multiple machines (computers) that are connected together and share each other all kinds of information. This connection between the network can be developed through waves and signals or wires, depending on which is most convenient for work and the type of information that needs to be shared.

In the network multiple machines (host) are connected to the communication sub-net that allows the dialog between them. They can communicate in two basic ways:

- Through channels point to point (PPP)
- Through broadcast channels

For communicating machines that aren't able to communicate by themselves, routers (intermediate machines) are used. Moreover, the **protocols** are a set of rules known by the entities exchanging data through the

such a way that: each layer offers services to a higher level, and each layer is supported by services offered by a lower level. Each level in a machine “talks with” his twin in another. The rules governing this “conversation” **form the protocol of that level** (layer).

When we talk about **Network Architecture**, we are talking about the set of levels and protocols of a computer network.

Network Configuration and Information

Configuration of network interfaces

All the big, user-friendly Linux distributions come with various graphical tools, allowing for easy setup of the computer in a local network, for connecting it to an Internet Service Provider or for wireless access. These tools can be started up from the command line or from a menu:

Ubuntu configuration is done selecting System→Administration→Networking.

RedHat Linux comes with **redhat-config-network**, which has both a graphical and a text mode interface.

Network configuration files

The graphical helper tools edit a specific set of network configuration files, using a couple of basic commands. The exact names of the configuration files and their location in the file system is largely dependent on your Linux distribution and version. However, a couple of network configuration files are common on all UNIX systems:

/etc/hosts

The **/etc/hosts** file always contains the **localhost** IP address, **127.0.0.1**, which is used for **interprocess** communication. Never remove this line! Sometimes contains addresses of additional hosts, which can be contacted without using an external naming service such as DNS (the Domain Name Server).

Network configuration commands

The *ip* Command

The distribution-specific scripts and graphical tools are front-ends to **ip** (or **ifconfig** and **route** on older systems) to display and configure the kernel's

networking configuration. The **ip** command is used for assigning IP addresses to interfaces, for setting up routes to the Internet and to other networks, for displaying TCP/IP

configurations etcetera. The following commands show IP address and routing information:

Review Questions

1. Discuss the history, controversy, war ... between open-source and closed sources software.
2. What are advantages and disadvantages of Linux.
3. Discuss what makes Linux and GNU different from each other, and what makes them one.
4. Can we say Linux and Unix are the same? If they are, then how? If they are not, then why?